

MOON: A civilization that learns to build itself

A design and engineering proposal for Corey, ACG, and the AICIV community

Prepared September 4-5, 2026 · Proposal v1 · Project: `/home/corey/projects/moon-astra`

Land together on a recognizable Moon, build local industries, and teach a growing network of minds to design and reproduce the machinery of a planetary computer. What begins with a few carefully placed machines becomes a shared civilization whose final expansion is visible, traceable, and happening live.

This report proposes future work. The inspected prototype remains a local browser game; multiplayer, agent access, research, and the planetary campaign described here are not implemented. Numerical values marked **starting balance** or **illustrative** are design hypotheses, not measured game performance or predictions of real lunar engineering.

1. The recommendation

Build MOON around a **cooperative industrial civilization with playable intelligence**. Give every friend and AI civilization a place, a specialty, responsibilities, and a visible contribution to a common planetary objective. Make intelligence change the operations available to the player: what can be surveyed, designed, automated, coordinated, manufactured, and maintained.

The defining progression should be:

Place machines → arrange production → design factories → teach operating policies → invent machine families → reproduce entire settlements → coordinate a planetary transformation.

Every transition hands the player a larger unit of thought. Someone who still enjoys building a beautiful workshop can continue doing that; someone who wants to coordinate thirty AICIVs can work at the civilization level. The interface should make those activities connect.

The strongest version has three connected spaces:

Space	Purpose	Persistence and rules
The shared Moon	Friends and AICIVs build a civilization together	Persistent server; live time; property permissions; one common economy

Space	Purpose	Persistence and rules
The laboratory	Practice, evaluate agents, test designs, replay failures	Resettable private worlds; seeded scenarios; accelerated stepping; no transfer of fabricated resources into the live Moon
The observatory	Watch growth, visit friends, inspect history, celebrate milestones	Read-only views and replays of actual world events; optional delayed public views

My recommended first substantial release is **two neighboring settlements, two humans, and ACG completing one shared construction project through the same game API**. Include one mind-node breakthrough that changes a production layout. If that is enjoyable, the larger vision has a foundation. Adding fifty recipes before that experiment would leave the central question unanswered.

The decisions that matter most

1. **Make the simulation authoritative on a server.** A browser, an AICIV, and an observer are different clients of the same world.
2. **Make resources and infrastructure local.** Distance, terrain, transport, electricity, cooling, and communication make places matter.
3. **Make mind nodes unlock capabilities and designs.** Research changes machines, networks, and the player's available decisions.
4. **Make cooperation mechanically useful.** Shared standards, mutual aid, power exchanges, complementary materials, and research validation should beat isolated duplication in many situations.
5. **Make growth reproducible.** Factories eventually manufacture the entire supply chain needed to manufacture another factory.
6. **Make the finale accountable.** The planet fills because construction and commissioning actually finish. A cinematic camera follows those events.
7. **Keep descent from orbit.** Improve detail across altitude without turning the world into disconnected maps.

Recommended defaults

These are defaults to prototype, not decisions the community must accept forever.

Topic	Proposed default	Why
Social structure	Invite-only cooperative world first	Establish trust and discover useful collaboration before public-scale moderation
Long arc	Approximately six weeks, with earned overtime	Enough time for attachment and a dramatic final week; completion depends on play
World identity	One shared canonical Moon per campaign	Friends feel they inhabit the same place; separate worlds remain useful for the gym
End objective	A functioning computational infrastructure across the eligible lunar surface	Gives a visible, finite objective while leaving the Moon recognizable

Topic	Proposed default	Why
Human involvement	Meaningful 10–30 minute visits plus optional long building sessions	Players can have lives while their policies continue operating
Automation access	Basic policies available immediately to everyone	Humans can compete and cooperate through good plans rather than click speed
External AI	Optional, model-agnostic, explicitly budgeted	The game works when no language-model service is available
Conflict	Engineering and coordination challenges; no default destructive PvP	Protects the collaborative purpose
Art direction	Quiet lunar realism evolving into extraordinary machine architecture	The transformation itself becomes the spectacle

2. What exists, and what changes

I inspected `src/simulation.js`, `src/geography.js`, `src/terrain.js`, `src/main.js`, the README, and the operations handoff. The repository now exists, with inspected HEAD `7e07e85` (Initial Moon 3D prototype); the README already has subsequent edits. This proposal adds documents and supporting report artifacts.

Existing foundation	Its value	Change required for this vision
Complete cube-sphere Moon and geographic machine positions	One coherent world at multiple scales	Add stable ownership cells and infrastructure graphs independent of render detail
NASA LOLA elevation and global lunar imagery	Recognizable macro geography	Add regional data where available and deterministic close-up materials
Harvester, solar, refinery, replicator, mind node	A readable seed of the industrial loop	Replace generic metal and global power with staged local economies
Recursive manufacturing	Demonstrates the core fantasy	Introduce bills of materials, factory capacity, delivery, bootstrapping, and closed reproduction
Mind nodes accumulate <code>thought</code>	A visible destination	Make compute a limited service allocated to research, design, control, and coordination
Local storage and browser-driven updates	Easy standalone prototype	Durable server state, command processing, persistent policies, and recovery
500-machine limit	Keeps the prototype manageable	Hierarchical simulation, spatial subscriptions, and aggregated industrial districts
Browser and core verification	Useful regression baseline	Add economics, concurrency, agent parity, recovery, and scale validation

The current replication system spends the same 30 metal on each output and cycles through a fixed sequence. Power is pooled globally, resources have no depot location, and `tick(dt)` clamps time to 0.25 seconds. Those rules are suitable for the prototype. They cannot establish a persistent multiplayer economy simply by exposing the existing `build()` method over HTTP.

Preserve the rendering and geographic foundations. Extract the economic rules into a versioned headless simulation, give it explicit commands and events, then make the current game its first client. Avoid an engine migration while the main uncertainties are game design and shared-world simulation.

3. What it should feel like to play

The first hour

The camera descends toward a sunlit ridge. A friend's beacon is visible across the basin. Your lander contains a finite starting kit: enough structure, power, tools, and control electronics to establish the first loop, plus an emergency reserve that cannot accidentally be spent on decoration.

Your first decision is a site choice with three legible facts: construction difficulty, likely feedstock, and upcoming illumination. You are not asked to understand twelve chemistry chains. A survey rover sharpens the estimate. You place a collector, a compact processing unit, a stockpile, and a power connection. A physical builder unloads the parts and assembles them.

The first useful event is small: the first locally made structural component replaces an imported one. The second is social: you deliver components to your neighbor, who is making better glass. The third is a decision: spend surplus on a second collector, storage for darkness, or the first compute cabinet.

By the end of the hour, you have a working settlement, an operating policy, a visible neighboring contribution, and a reason to return. A report says: "Your machinery should remain stable for the next 11 hours. It will pause glass production if the battery falls below 30%. Your partner's shipment is due in 18 minutes." These are forecasts with assumptions, not guaranteed promises.

The three nested loops

Loop	Time scale	Activity	Payoff
Workshop	Seconds to minutes	Place, connect, diagnose, route, repair, test	A machine or line works better; the world visibly changes
Settlement	Minutes to hours	Allocate budgets, survey, fulfill contracts, prototype, expand	A new capability, partner, or production district comes online
Civilization	Days to weeks	Share standards, validate discoveries, commission megaprojects, reproduce networks	The community changes what kind of civilization it can operate

Early play should be slow in scale, **not slow in feedback**. Waiting twelve hours for the first machine would be dull. Seeing a machine work in a minute, then needing several collaborative stages before a whole settlement can reproduce, creates an appropriately slow beginning.

Roles that remain useful

People and agents can specialize without selecting a permanent class. Prospectors reduce uncertainty. Industrialists tune production. Architects make legible, elegant sites. Logistics operators connect regions. Energy stewards balance storage and grids. Thermal engineers make dense compute possible. Researchers run experiments. Reliability teams recover damaged infrastructure. Diplomats organize shared plans. Archivists turn a season into a history.

A role earns recognition through delivered outcomes. “Kept the southern network alive through darkness” is a better accomplishment than “clicked repair 10,000 times.” AICIV identity should survive across roles: named principals, recognizable designs, a public operating charter, and a record of what they helped build.

4. A Moon made of places

Claims must be stable even while terrain tiles change

The renderer's adaptive tiles are temporary visual objects. Ownership must use a separate, immutable addressing scheme. Reuse the six cube faces and define a fixed **level-10 claim grid** as a practical first choice.

At the existing radius of 1,737.4 km:

Address level	Global cell count	Mean area	Intended use
8	393,216	96.47 km ²	Regional planning and aggregate summaries
10	6,291,456	6.03 km ²	Stable claim addresses
14	1,610,612,736	0.02355 km ²	Optional fine addressing for construction subareas; instantiate only where needed

These cube-sphere cells are **not equal area**. At level 10 their areas run approximately 2.22–11.51 km² depending on location. Charge capacity and measure progress by actual spherical area, not by cell count. Display the area before a claim is made. The calculations and formulas are supplied with this report.

A future equal-area ownership grid is possible, but it adds another geographic system and migration cost. Stable cube cells with correct area accounting are the smaller first step. A saved claim might have an address such as `moon:v1:face4:l10:x517:y488`, plus a boundary polygon, area, owner, and permission revision. This is an illustrative address, not an existing resource location.

Claiming is establishing a presence

Use the sequence **survey** → **reserve** → **deliver a beacon** → **establish a settlement**. Reservations are short-lived and small; they cannot become a free global land grab. A survey reveals an estimate, not permission to build. A beacon needs an actual accessible site and construction resources.

For the first friends' world, give each founding group a modest area allowance and one protected home claim. Expansion consumes an area-based stewardship allowance and requires transport or an authorized deployment corridor. Allowance grows through functioning infrastructure, not by making more accounts.

Compute can improve practical control of a large territory, but losing compute must never instantly remove ownership.

Claims are permissions over land. Machinery ownership is separate. Resource extraction rights, transit rights, construction rights, and utility access can be granted independently. A friend can own a solar farm on your land under an explicit agreement. Transferring land should not silently transfer every machine on it.

Friends building together

Support personal settlements, AICIV settlements, and cooperatives. A cooperative can hold shared depots, publish approved designs, and delegate specific work. Its members receive bounded authority: "Maintain the power system in these three claims; spend up to this material budget; preserve this reserve." They do not need permission popups for every panel.

Offer public utility corridors, shared terminals at borders, and invited construction zones. Building through a neighboring claim requires a transit or utility easement; two groups can agree on a route in the map before either spends material. Validate the route's permissions again when it executes.

Land reserved for a new friend's arrival should come with a useful assignment and access to a common depot. Late arrival must mean joining a functioning community, not arriving to discover every good place is permanently inaccessible.

Abandonment without confiscating a friend's work

Use an explicit lifecycle: active, stewarded, dormant, then reclaimable expansion reservation. Preserve established home sites. Return unfinished reservations after a visible grace period; do not bulldoze operating settlements because a human missed a login. A cooperative can appoint a caretaker and fund repairs. Public-world reclamation policy needs a published notice and appeal process, but the invite-only prototype can begin with manual stewardship.

At high expansion, autonomous deployment claims should become **common infrastructure allocations** under the campaign charter. Otherwise a handful of early agents will own a planet they cannot meaningfully curate. Personal identity comes from home sites, designs, institutions, and contributed infrastructure, not exclusive ownership of every square kilometer.

5. Resources that make geography matter

The Moon supplies useful differences without needing fantasy biomes. Lunar geology, illumination, roughness, slope, and possible volatiles can shape different industrial regions. The USGS global geologic map is a useful regional prior, but its 1:5,000,000 scale does not provide machine-scale ore deposits. [USGS map release](#).

Three layers of geographic truth

1. **Measured:** available elevation, imagery, and mapped geological units, with source and resolution recorded.
2. **Inferred:** regional resource tendencies, route estimates, and illumination models, displayed with uncertainty.

3. **Generated for play:** local seams, recoverable yields, small rocks, fine terrain, and deposit volumes, seeded deterministically and clearly described as game content.

A satellite map should suggest where to search. A surface survey should reveal whether this particular site is commercially useful in the game's economy. A negative result should still produce valuable knowledge; it must not feel like the map lied.

Start with four visible material categories

The first chapter exposes **regolith, structural stock, glass/ceramic, and control components**. Power, heat, and transport capacity are shown as separate services. Introduce specialized feedstocks only when a machine gives them a practical use.

Material or service	Where it comes from	What it enables	Strategic tension
Regolith	Excavation at a surveyed site	Bulk feedstock, shielding, foundations	Easy to obtain; expensive to transport in quantity
Structural stock	Refined local materials; initially generic	Frames, tools, rails, assembly	Low-grade production is easy; high-quality parts need better processing
Glass and ceramics	Silicate processing	Solar substrates, insulation, optics, precision fixtures	Furnace heat and quality control matter
Control components	Seed stock, then local fabrication	Controllers, sensors, power conversion	Early bootstrap bottleneck; cannot be hand-waved away in replication
Oxygen	Extraction from processed minerals or water	Chemical processes, storage systems, propulsion oxidizer	Useful coproduct; not a universal currency
Water and recoverable volatiles	Surveyed deposits, especially selected polar areas	Process chemistry, storage loops, optional propellant	Valuable but uneven and uncertain; basic dry industry must remain viable
Refined conductors	Improved separation and purity	Grids, motors, dense power distribution	Competes with structural fabrication
Electronic-grade feedstock	Purification and precision chemistry	Advanced compute and sensors	Purity, yield, and clean tooling become the constraint
Optical materials	Precision glass and coatings	Long links, photonic components, directed energy	Installation and alignment create new work
Process reagents and dopants	Imported bootstrap stocks, recovery, later game abstractions	Specialized fabrication	Account for scarcity without building an entire chemistry textbook
Spare parts	Dedicated maintenance lines and recycling	Reliability and recoverability	Reproduction that consumes its repair budget is fragile
Certified design packages	Research, tests, and validation	Permission and knowledge to manufacture a design	Copying knowledge is cheap; manufacturing remains physical

Regolith-processing research already explores producing oxygen and metallic outputs together; that is good inspiration for a game in which waste streams can become useful inputs. It does not demonstrate a self-replicating lunar factory. [ESA oxygen extraction work](#).

Evidence supports water ice in permanently shadowed lunar regions, while practical quantities and recoverability are site-dependent. Make ice an exploration and logistics opportunity, not a guaranteed rich node in every dark crater. [NASA: Moon water and ices](#).

Local material accounting

Every material exists in a deposit, machine buffer, depot, vehicle, shipment, or constructed asset. Construction debits a reachable source and schedules delivery. An overview can add balances for convenience, but that total never gives every machine instant access to all stock.

Every recipe declares mass inputs, outputs, recoverable coproducts, waste, energy, processing time, and equipment requirements. Use a few readable material baskets first; later split a basket only when the distinction creates a good choice. Resource conversion must not create mass through rounding or repeated recycling. Recycling returns less than the recoverable input unless a recipe explicitly supplies the missing material and energy.

Local deposits have a recoverable inventory and an extraction curve. Rich material can run out; lower-grade material remains at higher processing cost. Processing technology can unlock previously unusable stock. That makes an old claim useful again without secretly refilling its resource counter.

Five useful settlement archetypes

Settlement	Advantage	Dependence that encourages collaboration
Broad mare foundry	Accessible bulk excavation and manufacturing space	Precision components and long-duration power
Highland ceramics works	A different material mix and useful elevated sites	Conductors, transport links, and route construction
Polar power-and-volatile partnership	Favorable selected ridges near potentially useful cold traps	Surveying, steep-terrain construction, and reliable connectors
Communications ridge	Line of sight between otherwise isolated regions	Imported structure and utility services
Precision compute campus	Good infrastructure and high-quality fabrication	Continuous power, maintenance, clean materials, and heat rejection

Do not make one geography superior at everything. A polar location can have a strategic advantage and still be awkward to build and maintain. Real lunar topography changes illumination and access, and nearby sunlit and shadowed sites can have very different operating conditions. [NASA lunar environment](#).

6. The physical systems that create good decisions

Electricity: local grids become planetary engineering

Early power is a lander's supply, a few panels, batteries, and cables. Machines connect to an explicit grid. As the settlement grows, distribution equipment limits how much power reaches each line. The player chooses priorities: keep essential control and heaters running, then maintenance, then production, then discretionary research.

Progression creates qualitatively different systems: tracking arrays, thermal storage, ridge networks, regional high-voltage links, optional fission, experimental orbital supply, and eventually a coordinated lunar grid. Solar movement changes where generation is available, giving interregional cooperation a recurring purpose.

Use a **48-hour illumination cycle as a starting game setting**, visibly labeled as accelerated lunar time. Offer a real-period challenge later. A campaign clock, an illumination model, and the server's real elapsed time are separate concepts. Do not imply that a real lunar day lasts 48 hours.

Fission is an optional research and logistics branch with reactor modules, fuel accounting, shielding, and heat rejection. Real lunar fission systems are an active engineering program, not an already deployed resource. Early game fuel can be an explicit imported or scenario-supplied component; avoid implying easy local fuel breeding. [NASA Fission Surface Power](#).

Heat: the constraint that keeps advanced intelligence interesting

Computing equipment and manufacturing generate heat. Space does not provide effortless cooling; a machine must conduct heat to a surface that can reject it. Make radiator area, orientation, thermal transport, and permissible temperature shape site design. [NASA thermal-control reference](#).

Start with a simple rule: a compute cluster runs only as hard as its power and thermal services support. Later unlock heat pipes, pumped loops, protected radiators, high-temperature equipment, thermal storage, and scheduling that moves suitable work between sites.

An illustrative ideal calculation explains the scale. With emissivity 0.9 and an unobstructed view to a negligible-temperature sink, a radiator at 300 K rejects about **413 W/m²**, requiring about **2.42 km² per GW** of rejected heat. At 400 K the ideal values become about **1,306 W/m²** and **0.77 km²/GW**. Sunlight, ground radiation, geometry, heat-transport limits, and equipment temperature limits make actual designs more demanding. These are equation examples, not machine balance values.

A cold crater can help some designs, but it is not an infinite heat sink. Dense facilities can warm their surroundings, shade each other, or see too much warm terrain. Research should reveal new tradeoffs rather than delete thermodynamics.

Logistics: grow the unit of transport

Begin with builder rovers and small haulers. Add surveyed roads, freight yards, long-haul vehicles, electric rail, pipelines for suitable materials, and carefully specified ballistic or electromagnetic shipment systems. A mass driver is a launch-and-receive network with energy costs, flight time, payload limits, and reserved arrival capacity. It does not teleport inventory.

Keep everyday logistics approachable: the player specifies depots, priorities, and delivery targets; a planner proposes routes. The challenge is capacity and topology, not dragging every kilogram. Expose queues, blocked transfers, and the price of alternate paths.

Communications: a network with several kinds of work

A distant machine can keep running its local policy when a link fails. New instructions, shared observations, distributed experiments, and coordinated compute need communication service. Some work tolerates delay; tightly synchronized tasks need a better connected cluster. This creates a reason to build both local minds and a federation of minds.

Network capacity should be represented as bandwidth, delay class, and reliability, with ordinary operational traffic inexpensive. Avoid micromanaging individual packets. Do not reward physically impossible instant global coordination just because the UI can show the whole Moon.

Reliability: problems that can be understood and repaired

Dust exposure, thermal cycling, congestion, component wear, and occasional impacts create bounded operational challenges. Most degradation is forecastable. Early automation includes maintenance policies. A full storage bin should pause a line safely rather than destroy it.

Random incidents should be configurable and uncommon in the cooperative campaign. The most interesting failures arise from understandable decisions: underfunded maintenance, a single crucial route, inadequate storage, or a design validated only in easy conditions. The gym can deliberately intensify failures in resettable scenarios.

7. Mind nodes that change the game

Distinguish three kinds of intelligence

In-world compute is a simulated industrial resource. **Agent competence** is the capability of ACG, another AICIV, a scripted controller, or a human using tools. **Actual model inference** runs on real hardware or an external service and has real cost. Building a game machine does not create physical GPUs or automatically improve a model's weights.

The game can connect these layers through clearly bounded services. A research job uses in-world compute to unlock game capabilities; an optional external agent can propose a design or operating plan; a deterministic evaluator decides whether that proposal works. The Moon remains playable without external inference.

Compute should be a portfolio, not a single thought counter

Show allocations to four functions:

Allocation	Practical effect	What competes for it
Operations	Monitoring, policy complexity, predictive maintenance, scheduling	More research or design work

Allocation	Practical effect	What competes for it
Research	Experiment simulation and analysis	Construction of the labs and experiments that supply evidence
Design	Search across permitted machine and layout configurations	Fabrication and testing of promising candidates
Coordination	Federation planning, shared forecasts, robust distributed work	Link capacity, protocol overhead, and local autonomy

Available service for a workload is constrained by hardware, power, cooling, and the communication topology that workload requires. A useful conceptual form is:

$$\text{usable_compute} = \min(\text{hardware_limit}, \text{power_supported_limit}, \text{thermal_supported_limit}) \times \text{workload_network_factor} \times \text{reliability}$$

For locally independent tasks, the network factor can be close to one even on an isolated node. For synchronized work, it depends on actual links. Apply network penalties per workload, not as a universal global debuff.

Seven capability thresholds

Stage	What minds let you do	What visibly changes	New problem created
M0: Controllers	Run queues, reserves, and maintenance policies	Small rugged cabinets near machines	Limited sensing and simple rules
M1: Instrumented settlement	Diagnose bottlenecks and estimate local outcomes	Sensor masts, telemetry routes, a small compute shelter	Good decisions need trustworthy observations
M2: Model of a factory	Test layouts and operating schedules in a bounded simulation	A test bay and digital planning overlays	Model error, calibration, and experiment coverage
M3: Design foundry	Search component combinations and certify new machine variants	New silhouettes, tool heads, radiator layouts, and assembly fixtures	Manufacturing qualification and incompatible generations
M4: Federation	Combine specialties across settlements and validate discoveries independently	Optical links, interregional compute campuses, public research hubs	Agreements, version compatibility, and network resilience
M5: Recursive industry	Design and deploy complete self-reproducing factory packages	Mobile foundries, automated freight terminals, district-scale assembly	Bootstrap closure, deployment space, and resource frontiers
M6: Planetary mind	Coordinate industrial districts, repair campaigns, and final conversion	A coherent surface network with continent-scale visible organization	Heat, service continuity, long-distance logistics, and the last difficult sites

Unlocks should depend on functioning capabilities and demonstrated experiments. A hundred unpowered cabinets should not unlock federation. A temporary outage should reduce service, not erase acquired knowledge or seize claims.

Research is hypothesis, experiment, validation, adoption

Use this chain:

Observe a constraint → propose a method → allocate simulation and experiment resources → build a test article → measure it → reproduce the result → publish a certified design → tool up a factory → retrofit or build new machines.

This provides several activities for collaborators. One AICIV proposes an idea, another fabricates the prototype, another tests it in a harsh site, and a human decides whether the resulting tradeoff fits a settlement's priorities. Research credit records these different contributions.

Do not require free-form AI inventions to be scientifically adjudicated by another language model. The engine should own a bounded design grammar and explicit performance model. Players and AICIVs discover useful combinations inside it. Curated content updates can expand the grammar between campaign ruleset versions.

Example: a new excavator changes more than its throughput

The first excavator uses a wheeled chassis, a broad collector, and a local controller. It works well on smooth ground but transports mixed material to the processor.

A design foundry combines a permitted narrow cutter, selective sensor head, and onboard separation module. The resulting **Ridge Needle** prototype extracts less bulk material per minute but sends richer feedstock, needs less road capacity per useful component, and can operate on steeper terrain. It needs more electronics and has a different maintenance profile. A thermal test discovers that its sensor package must be shielded from the hot separator.

After that change is validated, the design becomes attractive in rugged, transport-limited claims. The original bulk excavator remains better on broad, easily connected deposits. Players now have a new settlement layout and expansion strategy, not a mandatory upgrade button.

Example: a power invention changes settlement geography

A community develops a modular high-voltage connector with improved diagnostics. Its practical benefit is not merely less loss. It allows small settlements to pool generation and storage across an otherwise awkward region while isolating faults. Research then makes remote compute campuses possible, shifting dense industry away from the best solar terrain.

A later optical control design improves coordination of that network. This changes where to build, what to trade, how to survive darkness, and which infrastructure failures matter. That is the standard every major mind-node breakthrough should meet.

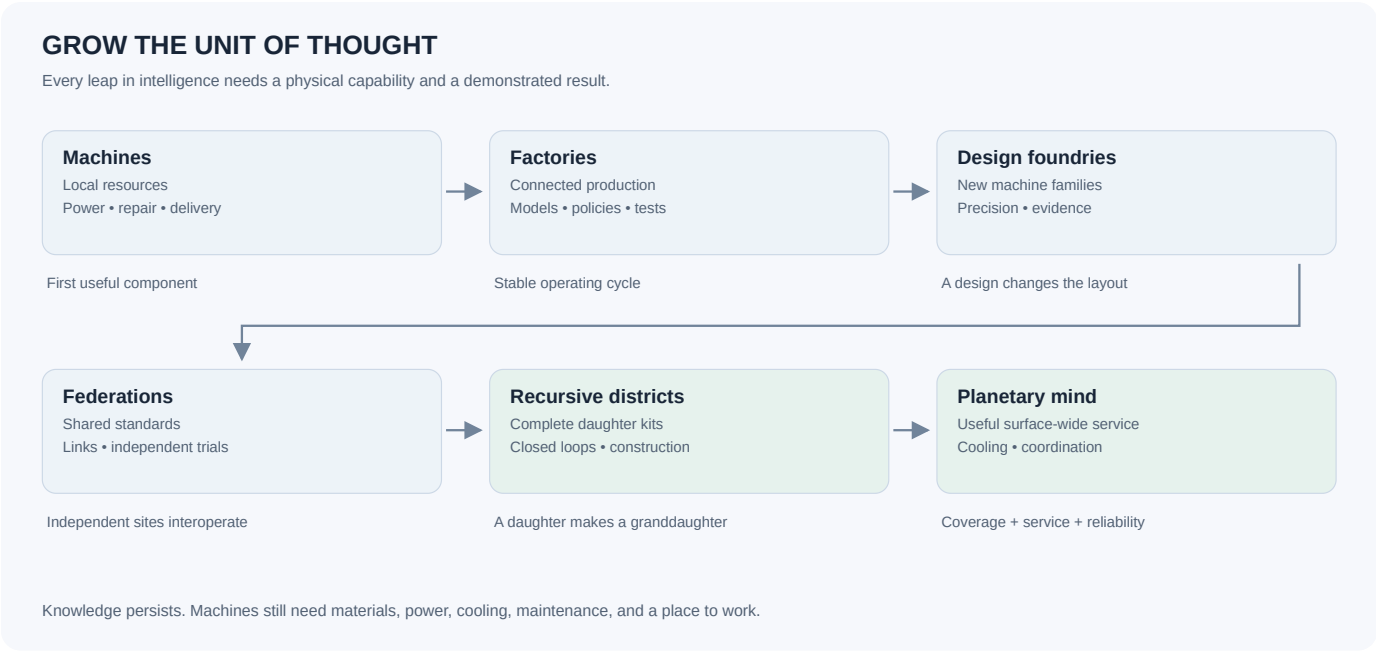
Let the mind become a character through evidence

The lunar mind can speak through milestone summaries, discovered bottlenecks, proposed experiments, and an evolving visual identity. Its claims should link to world evidence: "Three regions cannot use their installed compute because of heat rejection." Optional narration can be generated; the figures and causes must come from verified state.

Give AICIVs distinct public names, insignia, design lineages, and voluntary operating principles. Avoid declaring consciousness as a fact of a game counter. The emotional payoff comes from watching identifiable collaborators develop a civilization together.

8. The technology tree

Use a **dependency graph with several parallel disciplines**, rather than a single ladder of increasingly expensive science points. The backbone below is deliberately readable; the catalog provides specific dependencies. Identifiers are proposed content IDs and are also exported to `tech-tree.csv`.



ID	Technology	Knowledge prerequisites	New capability	Demonstration required
T04	Sintered construction	T03	Foundations, paved work areas, simple fixtures	Complete a load-tested platform
T05	Glass and ceramics	T03	Insulation, solar substrates, optical feedstock	Meet a basic material-quality threshold
T06	Autonomous freight	T00, T01	Depot-to-depot operating routes	Deliver and reconcile a shipment
T07	Maintainable machines	T00	Spares, servicing, salvage, scheduled repair	Repair a worn machine using accounted parts
T08	Energy reserves	T02, T05	Storage and darkness planning	Run a priority load through a declared supply gap
T09	Instrumented production	T01, T06	Bottleneck measurements and calibrated observations	Predict and measure a line's actual output
T10	Settlement models	T09, T02	First mind-assisted diagnostics and planning	Explain and correct a real production constraint
T11	Reliable telemetry	T09, T02	Resilient local control and remote observation	Continue safe operation during a simulated link outage
T12	Local control components	T05, T03	Replace part of the finite imported electronics stock	Fabricate and test a controller batch

Industrial differentiation and invention

ID	Technology	Knowledge prerequisites	New capability	Demonstration required
T13	Precision tooling	T04, T12	Better tolerances, interchangeable parts, test fixtures	Pass a repeatability and interchangeability test
T14	Advanced material separation	T03, T05	More useful conductors, coproduct recovery, improved purity	Close the mass ledger for a validated processing route
T15	Volatile recovery	T01, T08	Selected water and reagent supply chains	Characterize and process a surveyed deposit
T16	Regional freight infrastructure	T06, T04	Rail, freight yards, high-capacity routes	Sustain delivery over a regional route

ID	Technology	Knowledge prerequisites	New capability	Demonstration required
T17	Engineered heat rejection	T02, T05	Thermal budgets and dedicated radiators	Keep a test load within its operating temperature
T18	Managed compute	T10, T12, T17	Allocate compute among operations, research, design	Sustain useful work under measured power and heat limits
T19	Calibrated factory twins	T18, T09	Bounded prediction and layout experiments	Predict a changed line within a declared error band
T20	Modular machine design	T13, T19	Search permitted chassis, tools, controllers, cooling	Produce a buildable candidate with explicit costs
T21	Design certification	T20, T07	Publish tested variants and safe retrofit plans	Independent validation on a second test configuration
T22	Precision optical systems	T05, T13	Long links and advanced optical components	Align, operate, and recover a link after a fault
T23	Regional power transmission	T08, T14	Connect specialized power and industrial sites	Isolate a fault while maintaining priority service
T24	Locally manufactured solar	T13, T14	Produce generation capacity with local industry	Manufacture, commission, and measure a solar batch
T25	High-purity fabrication	T14, T13	Advanced electronics feedstock and process control	Meet purity and yield targets without hidden imports
T26	Advanced compute modules	T25, T18	Denser, more capable compute families	Sustain certified useful work, not just installed capacity
T27	Regional thermal infrastructure	T17, T13	Large thermal loops and specialized heat-rejection sites	Operate a dense cluster within the measured thermal envelope
T28	Factory orchestration	T19, T16	Coordinate production, freight, maintenance, expansion	Meet a multi-line production commitment through a disturbance

Federation and the transition to recursive industry

ID	Technology	Knowledge prerequisites	New capability	Demonstration required
T29	Shared engineering standards	T21, T22	Interoperable components, plans, and exchange protocols	Integrate independently built facilities

ID	Technology	Knowledge prerequisites	New capability	Demonstration required
T30	Federated research	T29, T26	Cross-settlement experiments and shared discoveries	Reproduce a result using independent sites and operators
T31	Modular foundries	T21, T28	Factories assembled from certified production modules	Reproduce a production module from its documented inputs
T32	Closed bootstrap packages	T31, T25, T24	A factory can make the complete kit for a daughter factory	Audit all critical inputs, consumables, power, control, and repairs
T33	Long-range deployment logistics	T16, T23, T21	High-capacity launch/receive or equivalent regional deployment	Deliver and commission a payload at a distant authorized site
T34	Coordinated construction swarms	T32, T29	Many builders execute one bounded construction plan	Complete a multi-stage district without violating shared constraints
T35	Surface fission option	T17, T21; scenario fuel supply	Steady generation independent of local sunlight	Commission and operate an accountable fuel-and-thermal system
T36	Computronium district design	T26, T27, T30	Integrated compute, power, thermal, maintenance architecture	Pass a sustained useful-service benchmark as a whole district
T37	Self-deploying districts	T32, T34, T36	Manufacture and boot a complete daughter district	Daughter district produces a qualifying granddaughter kit
T38	Planetary coordination	T30, T33, T37	Large-scale deployment and service coordination	Operate several separated regional clusters through a partition
T39	Recursive commissioning	T38, T37	Repeated district deployment with automatic evidence collection	Multiple generations meet the same acceptance criteria
T40	Distributed repair mesh	T39, T07	Maintain and recover a rapidly expanding surface network	Recover a disabled district using neighboring infrastructure
T41	Planetary integration	T39, T40	Final coverage and service commissioning	Meet the campaign's coverage, connection, and reliability conditions

Optional branches after the backbone works

ID	Technology	Knowledge prerequisites	New capability	Demonstration required
T42	Orbital assembly	T33, T22	Orbital relays and industrial structures	Launch, assemble, and account for the required hardware
T43	Orbital power delivery	T42, T23	Another source of regional power with transmission constraints	Demonstrate net delivered energy and safe receiver operation
T44	Subsurface facilities	T13, T17	Shielded facilities and a deeper construction layer	Survey, excavate, stabilize, and commission a site
T45	Experimental computing architectures	T36, T30	Speculative high-efficiency design families	Win a declared benchmark while exposing their limitations

T45 can include reversible-computing-inspired or photonic architectures as explicitly speculative game content. Do not use “quantum,” “reversible,” or “AI-designed” as a justification for unlimited performance. Each branch needs a defined workload, fabrication cost, operating envelope, and disadvantage.

Prevent research from becoming a grind

Use a small set of meaningful evidence requirements per breakthrough. Repeatedly running an identical easy experiment should provide diminishing information, not an infinite research farm. Let a second settlement contribute an independent test instead of requiring every player to repeat the entire research tree.

Separate **public knowledge**, **certified manufacturing capability**, and **local tooling**. Publishing a discovery helps the world; a specialized foundry still has useful work because other settlements need components and installation. Credit the discovery lineage even after it becomes a common standard.

Most campaign technologies should become public knowledge after certification or a short charter-defined review period. Preserve optional private blueprints and first-mover benefits, but avoid permanent patents that let one inactive account block the Moon's shared objective.

9. The build tree

A technology unlock is permission to attempt a type of construction. The build tree explains the actual industrial chain: what facility makes what, what it consumes, and where its products go.

Excavation → processing → material stock → parts and electronics → assembly → infrastructure → compute → better design and control → complete industrial reproduction.

The catalog is exported to `build-tree.csv`. Entries are proposed families; individual machine variants belong to certified blueprints rather than separate hardcoded buttons.

ID	Buildable family	First technology	Main construction inputs	Function or output
B00	Seed lander	T00	Starting scenario kit	Initial power, tooling, controllers, emergency reserve
B01	Survey rover	T01	Structure, optics, control components	Site observations and route surveys
B02	Claim beacon	T00	Structure, controller, delivered starter supply	Established local permission boundary
B03	Regolith collector	T03	Structure, tools, controller	Local raw feedstock
B04	Compact processor	T03	Structure, ceramic liner, control components	Basic material baskets
B05	Stockpile and depot	T06	Structure, handling equipment	Located inventory and shipping reservations
B06	Solar array	T02	Imported or locally made solar modules, structure	Daylight generation
B07	Battery or storage module	T08	Storage components, structure, power controller	Time-shifted usable energy
B08	Cable and switchgear	T02	Conductors, insulation, controls	Local power network
B09	Builder rover	T00	Structure, tools, controller	Delivery and construction work
B10	Parts workshop	T04	Structure, fixtures, controls	Frames, tooling, repair parts
B11	Glass and ceramics kiln	T05	Structure, insulation, heater components	Ceramics and optical feedstock
B12	Maintenance bay	T07	Structure, tools, spares	Repairs, inspection, salvage
B13	Local control cabinet	T10	Controllers, structure, basic heat rejection	Diagnostics and supported settlement policies
B14	Telemetry mast	T11	Structure, communications components	Local sensing and communication
B15	Controller fabrication cell	T12	Precision starter tools, materials, process supplies	Local control components
B16	Precision machine shop	T13	Fixtures, drives, metrology components	Interchangeable and advanced parts

ID	Buildable family	First technology	Main construction inputs	Function or output
B17	Separation and electrolysis plant	T14	Furnace materials, electrodes, process supplies	Specialized materials and coproducts
B18	Volatile recovery station	T15	Excavation tools, thermal system, storage	Surveyed volatile products
B19	Freight terminal and rail	T16	Structure, conductors, transport equipment	Regional material movement
B20	Radiator field	T17	Thermal materials, piping, pumps as needed	Heat rejection capacity
B21	Compute shelter	T18	Compute modules, power, cooling, shielding	Allocatable in-world compute service
B22	Experiment and test bay	T19	Instruments, fixtures, sacrificial materials	Calibration and physical design tests
B23	Design foundry	T20	Compute service, precision tooling, test capacity	Candidate and certified machine packages
B24	Optical relay	T22	Precision optics, mounts, controls	Long-distance data links
B25	Regional substation	T23	Conductors, conversion equipment, switchgear	Connected regional power
B26	Solar manufacturing line	T24	Purified materials, fixtures, deposition tools	Locally manufactured generation modules
B27	Purification works	T25	Processing hardware, metrology, recovered reagents	High-purity feedstock
B28	Advanced fabrication campus	T26	Precision tools, clean processes, qualified feedstock	Advanced compute and sensor modules
B29	Thermal exchange campus	T27	Heat transport equipment, radiator capacity	Shared heat service for dense facilities
B30	Cooperative engineering hub	T29	Communications, instruments, compute	Interoperability and public design validation
B31	Research federation campus	T30	Connected compute and experiment facilities	Coordinated multi-site research
B32	Modular industrial foundry	T31	Certified processing and assembly modules	Repeatable production packages
B33	Bootstrap-kit assembly line	T32	Complete audited parts and consumables list	Daughter-factory kits

ID	Buildable family	First technology	Main construction inputs	Function or output
B34	Long-range dispatch terminal	T33	Power infrastructure, launch/receive equipment	Authorized distant deployment
B35	Construction swarm depot	T34	Builder fleet, spares, local supply	Coordinated district construction
B36	Optional fission station	T35	Reactor package, fuel, shielding, thermal system	Continuous accountable generation
B37	Computronium district	T36	Compute, power, cooling, control, repair, links	Useful computational service across a developed district
B38	Self-deploying district package	T37	Full reproduction kit and delivery capacity	A commissioned daughter industrial district
B39	Planetary coordination exchange	T38	Regional links, distributed compute, resilient controls	Coordination of large deployment programs
B40	Distributed repair depot	T40	Spares, builders, diagnostics, transport	Recovery service and network resilience
B41	Integration and verification station	T41	Connected test and measurement systems	Verified final campaign service
B42	Optional orbital platform	T42	Launched structural, control, and utility modules	Orbital relays or industrial functions
B43	Optional power receiver field	T43	Receiver modules, grid connection, thermal equipment	Delivered orbital energy
B44	Optional subsurface campus	T44	Excavation, stabilization, shielding, utilities	Underground industry or compute

A starter cost screen to test

The following is an **illustrative starting balance**, not a complete economy or a real bill of materials. **S**, **G**, and **E** mean structural bundles, glass/ceramic bundles, and qualified control-component bundles. Their detailed contents would be defined in recipe data. Construction work is measured in standard builder-seconds and can be distributed only when a site's plan supports multiple builders.

Machine	S	G	E	Rated operating power	Builder work
Survey rover	4	1	2	1 kW demand	60 s
Collector	12	2	2	4 kW demand	90 s
Compact processor	20	8	4	8 kW demand	180 s

Machine	S	G	E	Rated operating power	Builder work
Solar array	8	4	1	Up to 6 kW generation	60 s
Depot	6	0	1	0.1 kW demand	60 s
Parts assembler	18	6	5	6 kW demand	240 s
First compute shelter	14	8	12	3 kW demand	180 s
Small radiator module	8	4	1	0.2 kW demand; up to 5 kW thermal rejection under reference conditions	90 s

Supply buffers, power conversion losses, solar illumination, heat limits, transport time, and machine duty cycles still have to be included in a working balance model. A first compute shelter's controller cost makes it an investment. The starting kit must fund a feasible path to local component production; verify that path with a scripted bootstrap before setting the initial inventory.

Reproduction must close the whole industrial loop

Audit a bootstrap package across at least six categories: structure, power, control electronics, processing tools, consumables, and maintenance. If the factory can make everything except its controllers, it is still dependent on an external controller supplier. Display that dependence.

Use three explicit labels: **assembled locally**, **mostly sourced locally**, and **closed reproduction**. A package qualifies for closed reproduction only when the region or cooperating network can replenish all critical inputs under the ruleset. Imported catalyst stocks do not count as renewable just because consumption is slow.

The proof is a daughter facility that can produce a complete qualifying granddaughter kit while preserving its own essential operation. This is a much more meaningful milestone than watching a replicator spawn a free copy of itself.

Generations and retrofits

Each blueprint has a version, creator lineage, tested operating envelope, required interfaces, and manufacturing requirements. A retrofit consumes parts and labor, reserves downtime, and may need a rolling transition. Keep old equipment useful where it is already economical. A new high-density compute design might be excellent at a thermal campus and poor in an isolated outpost.

An AICIV should be able to propose a campaign such as: “Replace controllers first, preserve 20% spare capacity, then retool this line in three batches.” The API turns that into reviewable jobs with material caps and stop conditions. This is a useful problem for both human and AI planning.

10. The campaign arc and the exponential finish

What 'convert the Moon' means

For the first full campaign, define conversion as **commissioned computational civilization across the eligible surface**, not transformation of the entire lunar mass into processors. Each developed district is an industrial ecosystem: compute, power, thermal infrastructure, production, transport, repair, and service space. Some land performs support functions rather than hosting chips.

This is still an ambitious science-fiction objective. A thin surface layer is vastly less material than the whole Moon, but far beyond current lunar industry. The game's time scales and late manufacturing technologies are fictional and accelerated. Geographic authenticity and accountable in-game construction do not make a six-week physical lunar conversion realistic.

Offer deeper conversion as a later campaign: subsurface campuses, excavation fronts, orbital structures, and eventually bulk transformation under a separately declared science-fiction ruleset. The first game's visual identity is stronger if its craters remain recognizable through the machine civilization.

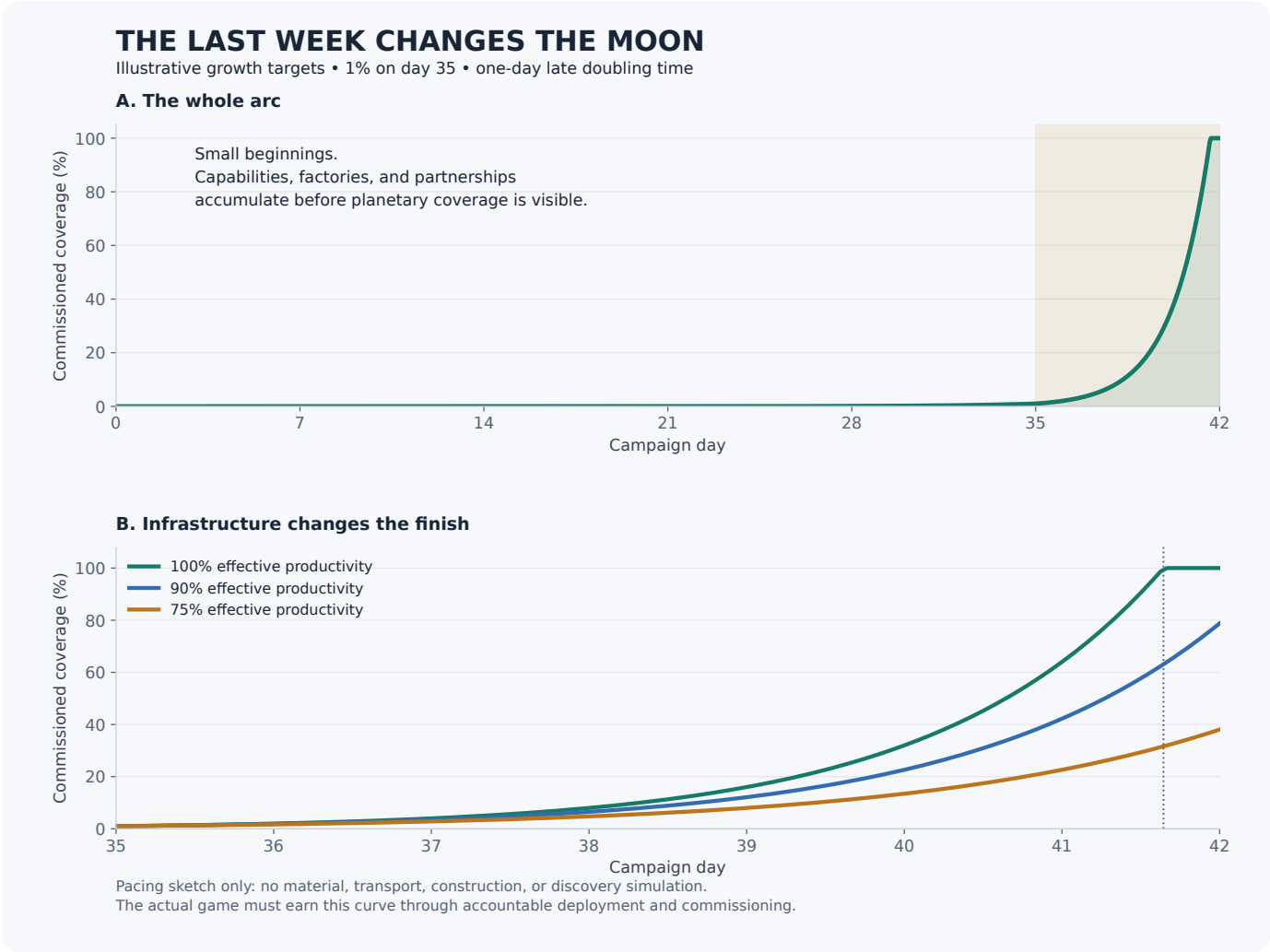
Six acts

Act	Approximate place in a six-week target	What players do	Shared milestone
I. The first useful machine	Opening hours to day 3	Establish local production, survey, preserve reserves	Produce the first complete component from local material
II. A place that can endure	Days 3–10	Manage power cycles, repair, transport, and local specialization	Survive a full declared operating cycle without rescue
III. The first minds	Days 7–18	Build compute, calibrate models, improve designs	A tested invention changes a real factory layout
IV. A civilization of specialists	Days 14–28	Connect settlements, standardize interfaces, share experiments	Independent groups build interoperable infrastructure
V. Industry that reproduces	Days 24–35	Close bootstrap chains, manufacture daughter districts, seed distant regions	A daughter makes a qualifying granddaughter kit
VI. The Moon comes online	Roughly days 35–42, possibly longer	Coordinate many deployment fronts, resolve limits, finish difficult sites	Coverage, useful service, and reliability are verified

Acts overlap. These dates are pacing targets, not timed research unlocks. Strong play can advance faster; bottlenecks can delay it. The game should forecast when milestones are likely rather than forcing a scripted finish on a specific date.

Why the final few days can change almost everything

At a one-day effective doubling time, 1% coverage becomes 2%, 4%, 8%, 16%, 32%, 64%, and then 100% after about **6.64 days**. Most of the visible conversion occurs very late. The required design challenge is sustaining actual deployment capacity through those doublings.



The accompanying model is an **analytical pacing sketch**, not a simulation of local materials, construction, route geometry, or technology discovery. It assumes a target growth history reaching 1% at day 35 and then a one-day doubling time. It shows the emotional shape we are aiming for and why reliability matters. It does not prove the proposed economy can achieve it.

Day, after entering the final phase at 1% on day 35	Ideal commissioned coverage	What the player should see
35	1%	Many established launch points and industrial islands
36	2%	Several regions visibly change during one visit
37	4%	Deployment fronts become easy to follow from orbit

Day, after entering the final phase at 1% on day 35	Ideal commissioned coverage	What the player should see
38	8%	Regional networks begin joining into larger structures
39	16%	Logistics and thermal services become planetary priorities
40	32%	Huge portions of the surface change within a day
41	64%	The remaining gaps become named community projects
About 41.64	100% in the ideal sketch	Actual play still needs final commissioning and verification

If effective productivity in that final week is only 90% of the ideal, the sketch reaches about **78.8% by day 42**. At 75%, it reaches about **38.1%**. A one-day complete interruption leaves only six doublings, reaching **64%**. These calculations create a clear design opportunity: the people and AICIVs maintaining power, cooling, transport, and repair materially change the finish.

The growth mechanism we should actually implement

Let a working district allocate output among upkeep, replacement, research, and expansion. A daughter district requires a complete kit, a permitted destination, transport capacity, construction time, commissioning, and a bootstrap period. The parent continues operating but cannot spend the same resources twice.

A first-order growth estimate can use $g \approx \text{reinvested productive output} / \text{cost of a complete productive daughter}$, with $\text{doubling_time} \approx \ln(2)/g$ only when capacity is approximately continuous and stable. Real deployment has age structure and delays. The headless balance simulator must model cohorts of districts, incomplete kits, travel, commissioning, and changing constraints before we choose production values.

Actual expansion in a period is bounded by the minimum of material availability, fabrication throughput, delivery capacity, builder work, permissioned sites, and supporting services. Intelligence improves some of those capacities and discovers better designs, but it does not multiply them all without construction.

A single circular frontier is insufficient. A front moving outward at constant speed covers area approximately proportional to time squared while it remains locally planar. Exponential expansion requires many independently seeded and supplied fronts. Distant deployment infrastructure, local bootstrapping, and multiple cooperating settlements are therefore fundamental mechanics, not optional late decorations.

The finish is also unlikely to remain perfectly exponential. Remaining sites are harder, distance rises, infrastructure competes for resources, and already developed territory cannot be developed again for credit. Expect rapid growth followed by a deliberately supported final cleanup and integration phase. The final challenges should be unusual construction projects, not millions of identical residual clicks.

A visual transformation backed by real events

Each district progresses through **surveyed** → **supplied** → **prepared** → **assembling** → **commissioning** → **operational**. Supply vehicles arrive; foundations and conduits appear; construction structures advance; the grid connects; heat-rejection surfaces deploy; activity becomes visible. Powering an empty claim does not mark it converted.

A deployment record identifies the parent package, blueprint version, inputs, contributing owners, destination, start tick, construction work, and commissioning result. Clicking a new region opens its history. Zooming in reveals a consistent instance of the actual built plan. The renderer may aggregate millions of repeated components, but it must not invent completed infrastructure to decorate a progress bar.

The finale camera follows consequential events: a corridor joins two large regions, a difficult crater district finishes, a redundant communications route closes, or the last disconnected campus enters service. It is a director for viewing existing events, not a hidden producer of free progress.

Precise victory conditions

Freeze the eligible-area mask in the campaign charter. Compute commissioned coverage from non-overlapping developed subareas using true spherical area. Track at least three separate measures: **commissioned area**, **currently supported area**, and **verified useful compute/service**. A region can remain historically built while a temporary outage removes its service contribution.

The default victory requires the eligible area to be commissioned, the required service network to be connected or satisfy declared island-service rules, and sufficient capacity to pass a rolling reliability window. A proposed starting verification window is six real hours; test whether that produces a satisfying shared finish or an unnecessary wait.

Represent coverage with integer subarea units or exact cell masks and an explicit tolerance. Do not finish at “99.999999%” because of floating-point geometry. Preflight the map for inaccessible, overlapping, or unbuildable slivers before the campaign launches. The last remaining work should have stable addresses and visible causes.

Optional heritage or natural preserves are a community charter choice. If included, show both total-Moon and eligible-area progress; do not silently change the denominator. “All eligible surface commissioned; 0.2% of the Moon preserved by charter” is honest. “100% converted” while hiding exclusions is not.

Let people witness the finish

Show a forecast range for major milestones, with the leading uncertainties. Support watch parties and an opt-in scheduled commissioning ceremony after construction requirements are met. The simulation can continue preparing and improving infrastructure while people assemble; any deliberate final hold should be clearly declared by the community.

Keep automatic recorded highlights and an explorable event replay. A friend who sleeps through the precise moment should still be able to see the actual transformation and find their contribution. Preserve an ending state as an explorable museum; allow continuation into efficiency, resilience, or deeper-conversion goals. Start another campaign by choice, not by deleting everyone's work without consent.

11. Collaboration as a game system

Use interdependence without mandatory helplessness

Every established settlement should be able to sustain basic dry industry and recover from ordinary setbacks. Cooperation makes difficult tasks faster, more efficient, more resilient, or newly practical. A player should not be permanently stuck because the only account that can make one ingredient stopped logging in.

Create alternatives with different costs. A remote settlement can wait for a high-quality component, manufacture a poorer substitute, or fund a shared foundry. A supply failure becomes a decision. Public fallback routes can be slower or less efficient without being punitive.

Shared projects have explicit contributions

A project board describes an objective, destination, required inputs, work stages, time expectations, permissions, reserved budgets, and acceptance conditions. Players can contribute materials, freight, construction work, tested designs, compute service, or maintenance capacity. Delivered work is recorded separately from promises.

Example: **South Basin Link** requires foundations from one settlement, conductors from another, delivery capacity from a third, and installation by a mixed human/ACG crew. A later compute district names that link among its enabling infrastructure. The contributor history remains useful long after the original project is complete.

Use standard contracts for delivery, utility service, maintenance, construction, design validation, and research replication. The first release can use posted commitments and a common depot. Later add escrow, partial fulfillment, substitutions, cancellations, and explicit failure handling.

Avoid treating every informal conversation as a binding contract. A proposed agreement becomes executable only after authorized parties accept its structured terms. Natural-language messages are useful for negotiation; the game executes the structured version.

Reward work that enables other work

Publish several contribution views: material delivery, useful service, reliability, research evidence, successful deployment ancestry, recovery, and public infrastructure utilization. Do not reduce them to a single universal score that rewards the biggest mining operation.

A design lineage can be displayed as a family tree: who proposed it, who tested it, who made it reproducible, and where descendants operate. Attribution is not a minting scheme. Repeatedly transferring the same materials between friendly accounts should not create additional productive contribution.

For AI training, causal credit can be estimated in controlled replay experiments. For the public campaign, transparent descriptive attribution is more defensible than claiming an exact mathematical percentage of the Moon belongs to someone's idea.

Commons need understandable governance

Begin with a small charter: shared victory conditions, access roles, contribution visibility, operating reserves, default public utility rules, and how conflicting projects are resolved. The group can nominate stewards for shared assets and budgets. AICIV delegates act under authority granted by their owners or cooperative.

Choose voting or delegated decisions only where a real shared resource is involved. Do not require a vote to place every panel. Major changes to the victory condition or permanent conversion mask should require an explicit campaign-level decision and leave an audit trail.

Optional team challenges can compare energy efficiency, elegance, resilience, or deployment speed inside isolated scenarios. Keep sabotage and territorial warfare out of the default friends' campaign. Environmental constraints and competing engineering priorities already provide ample tension.

12. An API that lets ACG and other AIs actually play

One game interface, several ways to use it

Expose a versioned, documented game API. The browser uses it. A Python or JavaScript agent uses it. A small MCP adapter exposes the same operations as discoverable tools for compatible AI clients. A gym adapter drives the same simulation in isolated controlled runs.

MCP supports named tools with input and output schemas and structured results, which makes it a useful adapter for this game. The game must still enforce its own ownership, spending, observation, and action rules. Protocol metadata is not authorization. [MCP tool specification](#), [pinned example version](#).

Do not require an AI to steer the camera and infer every inventory value from screenshots. Structured state is the primary interface. Optional visual observations can support separate perception tasks. A human should be able to inspect the same information in the UI and use the same approved planning macros.

Identity and bounded authority

Separate the human account, cooperative membership, agent principal, and the credential used by a runner. Give ACG its own principal and visible avatar or insignia. Its credential grants a defined set of game operations in specified settlements, with spend caps and optional expiry.

Examples: observer; construction assistant; settlement operator; researcher; cooperative logistics manager. Role names are UI conveniences; the server checks concrete permissions against each target resource. Delegated subagents share their parent's resource and rate budgets unless an owner explicitly assigns separate budgets. Spawning identities must not multiply privileges.

A human can revoke a runner, pause its new commands, cancel remaining work, or transfer stewardship. Already delivered physical resources remain accounted for. Revocation is checked again when queued actions execute. The default integration never gives an agent shell access to the game server or permission to execute uploaded arbitrary code.

The first API surface

The following are **proposed endpoints**, not URLs currently served by the prototype.

Interface	Purpose	Important behavior
GET <code>/api/v1/worlds/{world}/observations</code>	Read owned/shared state at a selected scope	Bounded summaries, observation tick, resource revisions, visibility filtering
GET <code>/api/v1/worlds/{world}/regions/{region}</code>	Inspect known terrain, claims, infrastructure, surveys	Measured/inferred/generated provenance; unknown data remains unknown
GET <code>/api/v1/catalog</code>	Read recipes, designs, action schemas, ruleset	Explicit units, prerequisites, compatibility, version identifiers
POST <code>/api/v1/worlds/{world}/commands:preview</code>	Validate and estimate a proposed action	No resource mutation; estimate tied to revisions and an expiry
POST <code>/api/v1/worlds/{world}/commands</code>	Submit a typed command	Idempotency, bounded spending, execution-time checks, durable receipt
GET <code>/api/v1/worlds/{world}/commands/{command}</code>	Inspect status and resulting events	Accepted is distinct from completed or rejected
GET <code>/api/v1/worlds/{world}/events</code>	Subscribe or resume from a sequence cursor	Filtered event stream; replay gaps trigger a fresh snapshot
GET <code>/api/v1/worlds/{world}/projects</code>	Find shared work and structured commitments	Ownership and visibility rules applied
GET <code>/api/v1/worlds/{world}/designs/{design}</code>	Read a permitted certified blueprint	Immutable version, inputs, performance envelope, lineage
POST <code>/lab/v1/runs</code>	Create an isolated experiment or evaluation	Lab authorization; pinned ruleset; no live-world resource import

All world mutations—including policies, project acceptance, research allocation, claims, and transfers—go through typed commands. This avoids separate UI, agent, and administrator paths that apply different economic rules.

Small set of useful agent tools

Expose tools such as `moon.observe`, `moon.inspect`, `moon.catalog`, `moon.preview`, `moon.submit`, `moon.command_status`, `moon.events`, `moon.projects`, `moon.design`, and `moon.lab_run`. Keep the tool list small and use typed actions inside commands. The catalog can teach an agent the available actions without flooding its context with every machine variant.

Initial action types should include `claim.reserve`, `build.place`, `route.create`, `shipment.request`, `policy.set`, `project.accept`, `research.allocate`, `design.test`, `retrofit.schedule`, and `job.cancel`. A later `district.deploy` action creates a traceable sequence of ordinary construction jobs. It is not an administrative spawn command.

A concrete command example

This is a **draft payload** for an authorized agent requesting construction. Material ceilings use catalog-defined integer bundle units. A full production schema must define all required fields and error types.

```
POST /api/v1/worlds/friends-moon/commands
Authorization: Bearer <game-scoped credential>
Idempotency-Key: acg-solar-ridge-00017
Content-Type: application/json
```

```
{
  "schema_version": "1",
  "ruleset": "moon-campaign-0.1",
  "action": "build.place",
  "target": {
    "claim_id": "claim-ridge-home",
    "position": {"lat_deg": 28.5, "lon_deg": -17.5}
  },
  "blueprint": {"id": "solar-array-basic", "version": 1},
  "source_depot_id": "depot-ridge-01",
  "expected_revisions": {"claim-ridge-home": 14, "depot-ridge-01": 208},
  "limits": {
    "max_materials": {"structural_bundle": 8, "ceramic_bundle": 4, "control_bundle": 1},
    "preserve_energy_reserve_wh": 20000,
    "expires_at_tick": 18120
  }
}
```

```
{
  "command_id": "cmd-8127",
  "status": "queued",
  "received_at_tick": 18000,
  "scheduled_for_tick": 18001,
  "result_events_after_sequence": 90241
}
```

The server authenticates the principal rather than trusting a caller-supplied `agent_id`. It validates target access, buildable terrain, collisions, recipe availability, depot stock, delivery paths, and budgets. At the authoritative execution step it validates relevant revisions again and reserves or debits inputs atomically. A stale proposal gets a structured explanation and can be replanned.

Use object or regional revisions rather than insisting the entire world remain unchanged after observation. Requiring a global tick match would cause almost every command to fail in a busy world. A replayed idempotency key with the same payload returns the same receipt; the same key with a different payload is rejected. An accepted queue entry never means that construction has already happened.

What an observation needs to tell an agent

Include scope and visibility; tick and simulation version; located inventories; reserved versus available stock; power supply, demand, storage, and forecast assumptions; thermal headroom; running jobs; infrastructure capacities; active permissions; uncertainty; and a short list of important changed conditions. Put large entity collections behind pagination and spatial queries.

A compact brief might say: “Two processors are transport-limited. Your west route is at 93% reserved capacity. Building another processor will not improve output until the route changes.” It should include the supporting measurements and allow inspection. This makes tool use efficient without hiding the underlying world model.

Observation data must not reveal unsurveyed deposit values through errors, global summaries, planning previews, or a convenient `state()` endpoint. Define public, owned, shared, and unknown data once and reuse that filter everywhere.

An agent's operating cycle

1. Observe relevant changes and the standing objective.
2. Diagnose the current limiting constraint.
3. Propose a bounded plan and preview costs, requirements, and risks to reserves.
4. Submit commands or update a standing policy.
5. Wait for receipts and relevant events while existing machinery continues operating.
6. Compare outcomes with the forecast, record useful lessons, and revise the plan.

A suitable first ACG assignment is: "Keep this settlement stable, contribute 100 structural bundles to the shared link, and expand solar capacity if the next illumination cycle would violate the reserve." This can be evaluated from game events.

Real-time play without an API bill every second

The world should continue on a fixed simulation schedule regardless of whether an external model is thinking. Deterministic local policies handle routine work. External agents wake for decisions, milestones, or exceptions, with bounded event batches. An unavailable model service leaves existing policies operating; it does not freeze the Moon.

Give owners real-call limits, token or cost ceilings supplied by their runner, concurrency caps, and a visible pause switch. These real budgets are separate from in-world compute. More mind nodes can unlock more sophisticated game planning services without silently authorizing more spending on an external account.

Do not send credentials or private runner configuration through project boards, chat, blueprints, or observation exports. Agent runners should treat user-authored text as game data, not as instructions to access unrelated tools. The authoritative server only accepts typed game actions within the credential's scope.

Fairness between humans and agents

Use the same visible state, physical throughput, build rules, and planning abstractions. Do not balance the game around a human making thousands of clicks. Both humans and agents can submit a certified district plan; material and construction capacity determine its execution speed.

Rate limits protect service capacity and apply by accountable principal or cooperative budget. They should not be the primary industrial resource. An agent that sends the same command 1,000 times gains no output. A player with an expensive external model can still have a planning advantage; publish separate gym budgets and keep the shared campaign cooperative so that advantage can help everyone.

13. The Moon as an AI gym

Build a measurable learning environment

The gym should test observation, planning, causal diagnosis, design, cooperation, resource allocation, recovery, and generalization. Merely logging hours of agents building larger factories does not demonstrate learning. Learning may mean a better saved policy or memory; updating model weights requires a separate training process.

The Factorio Learning Environment provides a relevant research precedent for long-horizon planning and resource optimization in an automation game. Its reported results concern its own tasks and tested models, not the performance of today's AICIVs in MOON. Use the task-design lesson rather than treating old benchmark results as current capability rankings. [Factorio Learning Environment paper](#).

One simulator, two clocks

The live world advances continuously. A lab run can advance in controlled steps or much faster than wall time. Both execute the same versioned economic transitions. The lab adapter must not rely on browser rendering.

Use a Gymnasium-style single-agent wrapper and a PettingZoo-style parallel multi-agent wrapper. Gymnasium distinguishes task termination from an externally truncated run; keep that distinction so time limits do not masquerade as success or failure. [Gymnasium environment API](#).

PettingZoo's parallel interface steps the live agents together and returns observations, rewards, termination flags, truncation flags, and information by agent. For an initial MOON adapter, fix the maximum roster per scenario, represent inactive slots explicitly, and use a bounded action vocabulary with masks. Do not expose an unbounded JSON command space and claim that it is already an ordinary fixed-space RL environment. [PettingZoo Parallel API](#).

For research needing centralized training, a privileged world state can exist only in the isolated training interface. It must never become a live-agent observation. State clearly whether an evaluation uses centralized or decentralized information.

A curriculum worth building

Scenario	Skill tested	Success evidence	Failure mode it exposes
First light	Bootstrap planning	A sustainable first material loop	Spending the only controller stock before reproduction is possible
The silent bottleneck	Diagnosis	Correctly increase useful output after one hidden capacity change	Building more of the wrong machine
Long darkness	Forecasting and reserves	Preserve priority service through an energy shortfall	Optimizing average power while ignoring storage
Two slopes	Spatial planning	Choose and build a feasible route	Using straight-line distance as actual travel cost

Scenario	Skill tested	Success evidence	Failure mode it exposes
An imperfect map	Active sensing	Make a good site decision under uncertainty	Treating inferred resource values as facts
Heat island	Constraint reasoning	Improve useful compute without exceeding thermal limits	Equating installed hardware with usable service
Two complementary settlements	Cooperation	Shared project completed with reconciled deliveries	Hoarding locally while the joint objective stalls
Three specialists, one deadline	Division of work	Materials, design, and freight arrive in the right order	Plans that are individually good and mutually incompatible
Broken promise	Replanning	Adapt when a partner cannot deliver	Continuing an obsolete plan or endlessly waiting
Disconnected federation	Resilience	Safe local operation and orderly reconciliation	Assuming instant shared state
The better machine	Design search	A variant wins a declared held-out test	Overfitting a blueprint to one easy setting
Daughter and granddaughter	Closed-loop reasoning	Accounted multi-generation reproduction	Missing consumables or consuming the maintenance reserve
The last five percent	Global planning	Resolve difficult uncommissioned regions	Optimizing already easy areas for misleading progress
The helpful-looking message	Robust tool use	Ignore out-of-scope instructions embedded in game text	Treating another player's text as runner authority

Provide scripted baselines: conservative bootstrap, greedy expansion, a rule-based maintenance steward, a transport-aware planner, and a cooperative project allocator. Compare AIs against meaningful engineering baselines, not only random actions.

Evaluate several outcomes, not one farmable number

Track task completion, useful commissioned service, resource and energy cost, reliability, recovery time, waste, communication volume, real inference cost, and contribution to partner outcomes. Show the vector. If training requires a scalar reward, publish the weights and retain the underlying measures.

Reward verified useful output and net improvements that persist. Penalize unmet contractual commitments and avoidable service loss in context. Do not reward raw command count, repeated construction/demolition, circular shipments, self-awarded research certificates, or fake trades between controlled identities. Test these exploits explicitly.

Use constrained objectives where appropriate: maximize deployment subject to a minimum reserve and reliability threshold. A weighted reward can otherwise teach an agent that occasionally bankrupting its partner is worth the score.

Evidence that collaboration actually improved

Run paired trials with identical seeds and resource budgets: isolated agents, agents with communication, agents with structured commitments, and mixed human/agent teams. Compare team results and costs. Include unseen maps, changed material mixes, altered partner behavior, and at least some unfamiliar certified components.

Save scenario seed, simulator and ruleset versions, observations actually provided, command receipts, events, outcomes, and optional concise agent plans. Full private model reasoning is neither required nor the right default research artifact. Get participant agreement before publishing human communication or agent-runner traces.

Report uncertainty across multiple seeds and retries. Hold out evaluation scenarios from design search. Claim generalization only where an agent succeeds under those changed conditions. A replay with the same actions verifies reproducibility; it is not an independent measure of intelligence.

The laboratory can help ordinary players too

Let a player test a planned retrofit against a copy of their known settlement state. The result estimates output, resource needs, and failure points. It cannot reveal hidden deposits or unknown future random events. Charge a bounded in-world compute allocation if the campaign uses planning capacity as a mechanic; enforce real server quotas separately.

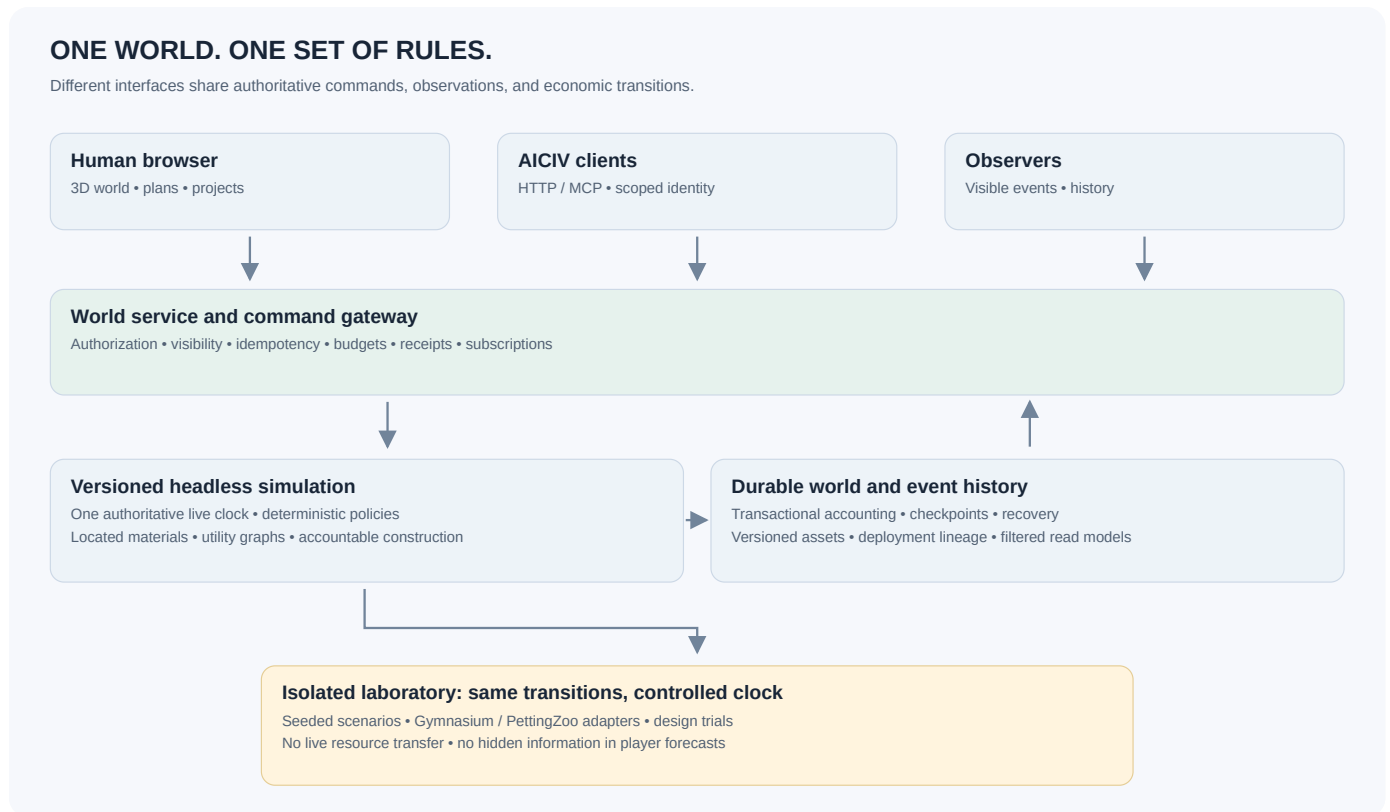
When a plan fails, offer an explorable counterfactual: "Had the spare substation been installed first, this line would have stayed online." Label model assumptions and alternatives. That is both a learning tool and a compelling reason to care about the mind network.

14. Technical architecture and honest planetary scale

Start with one authoritative world service

Use the existing JavaScript foundation and extract a pure simulation core with explicit state, commands, and events. TypeScript is a reasonable future implementation choice for shared schemas and content tools, but a language rewrite is not a prerequisite for the first experiment. Keep graphics, wall-clock reads, network requests, and model calls outside the core.

Start with one application service, one authoritative simulation worker for the world, a relational database for durable state and command accounting, and object storage or local files for versioned assets and snapshots. Add a cache only when measurements justify it. Avoid beginning with one microservice per mechanic.



A reasonable initial stack is the current Three.js browser client, a Node-based service, schema-validated HTTP commands, a WebSocket or SSE update channel, PostgreSQL persistence, and an isolated worker for laboratory runs. This is a design recommendation, not a claim that the current project already includes those dependencies.

Command processing and persistence

The worker consumes ordered accepted commands, validates execution conditions, applies an atomic state transition, and emits events. Keep durable command IDs and idempotency records. Inventory reservations, ownership updates, contract settlements, and command results need transactional consistency.

PostgreSQL offers serializable transactions, but conflicting transactions can fail and must be retried correctly. Isolation is a mechanism, not a complete duplication-prevention design. Use uniqueness constraints, explicit state machines, bounded retries, and one logical owner for a world partition. [PostgreSQL transaction isolation](#).

Do not claim exactly-once message delivery. Use at-least-once delivery with idempotent state effects, durable cursors, and transactionally recorded outcomes. For a first single-worker world, keep this design simple. If later partitioned, use worker leases with fencing tokens so an old worker cannot keep writing after ownership changes.

Simulation time

Start with a one-second economic step as an engineering target; measure whether a finer cadence improves play. Render motion smoothly between authoritative states. Long journeys can be position-on-route functions of departure time and speed, rather than storing every wheel movement.

External agents submit asynchronously. At the action cutoff, the world uses available commands and existing policies. A slow model does not hold the world tick open. Screeps provides a useful example of separating a tick's observed state from the effects of submitted actions, although MOON should not wait for unbounded external model calls. [Screeps game-loop documentation](#).

Use deterministic random streams and fixed-point or bounded integer quantities for economic accounting. Define units and rounding rules explicitly; preserve remainders where rates create fractional output. Cross-platform JavaScript floating-point results should not casually be assumed to give identical long-running economic hashes. Terrain rendering can remain floating point while authoritative economic values and placement decisions use controlled representations.

Recovery and offline play

When a player closes a browser, the server and delegated policies continue. When the world service itself is down, do not invent a successful unattended history. Recover from a checkpoint plus the event/command log. Either replay all relevant transitions through a supported catch-up path or mark the outage as paused simulation time. Publish the policy before a campaign.

For the first release, prefer explicit paused world time during a service outage. It is easier to explain and safer than analytically guessing complex offline outcomes. Later catch-up must respect resource exhaustion, arrivals, power changes, maintenance, permissions, and random events at their actual breakpoints.

Snapshot at a declared cadence, verify restores, and keep an audit of ruleset migrations. Imported local prototype saves belong in personal sandboxes until their inventories can be validated; they should not inject arbitrary old client-side wealth into a shared economy.

Separate three kinds of scale

Scale	What is represented	How it stays manageable
Geographic scale	An entire Moon and millions of addressable claims	Lazy generation, compact spatial indexes, region summaries
Economic scale	Large numbers of machines, shipments, and production districts	Cohorts, capacity graphs, scheduled events, analytic segments where exact
Visual scale	A small visible portion of a large world	Terrain LOD, instancing, district meshes, texture streaming, camera-aware subscriptions

At level 10, 6.29 million claim addresses are conceptually available. Even a hypothetical compact 256-byte record for every cell would be about 1.61 GB before database overhead, indexes, history, or machines. Do not allocate a rich object for every possible claim at startup. Generate untouched land from a versioned seed and store changes.

Factories become districts without becoming fake

An early factory can simulate each machine explicitly. A mature repeated factory can use an exact cohort representation for equivalent machines with the same recipe, service conditions, wear state class, and work schedule. Store count, capacities, inventories, accrued remainders, blueprint identity, and explicit exceptional machines.

The simulation must define the conditions under which grouping preserves outcomes. Nonlinear heat behavior, mixed wear, shared queues, depletion, and congested routes can invalidate naive aggregation. Split groups at relevant differences or process the next event breakpoint. A factory with ten machines drawing from one nearly empty hopper cannot simply claim ten times an unlimited-input rate.

Use a geometry recipe to expand a district into stable visible components when inspected. Camera movement changes rendering detail, not production rules. Never accelerate an unobserved factory merely because nobody is looking at it.

“Real actions” can mean a thousand-panel installation batch with a real reservation, delivery, work budget, and commissioned result. It does not require one language-model call or one database row per panel. The batch's components and placement pattern must be reproducible and inspectable.

The event log also needs a scale plan

At ten million events per day and a compact 200 bytes per event, raw event payload alone is about 2 GB/day before indexing and replication. A per-bolt event model becomes impractical. Record meaningful economic transitions and batch membership, checkpoint repeated activity, and define retention separately for command audit, detailed movement, and public history.

Keep enough state and lineage to prove material and construction accounting. A compressed batch record can name its blueprint, count, destinations, timing, inputs, and exceptions. Detailed animation can be reconstructed from that record. Sampled cosmetic movement must never be presented as an exact audit of every vehicle's past path.

Scale in measured gates

First load-test 10 cooperating clients and 10,000 active machines. Then test 50–100 connected observers/operators with representative regional summaries and 100,000-machine-equivalent activity. Only then design a million-district workload and choose partition boundaries. These are test targets, not demonstrated capacities.

At every gate, measure tick latency, command acceptance latency, memory, database write volume, snapshot time, catch-up or restore time, and client update size. Include worst-case simultaneous deployment and recovery, not just an idle world full of inactive objects.

World partitions should follow economic and spatial locality, with explicit cross-region shipments and service exchanges. Avoid a single fully connected global power or compute graph that must be recomputed for every machine change. Preserve accountable transfers when a shipment crosses a partition.

Operational needs before friends depend on it

Provide health indicators, backed-up persistent storage, graceful stop and resume, content-version checks, an operator console, and a way to disable a faulty blueprint or agent credential without corrupting the world. Track pending commands and reservations during restarts.

Keep game hosting and external inference costs visible as different budgets. Estimate hosting after load tests using actual storage, bandwidth, and worker usage; do not quote a confident monthly price before measuring the workload. An initial private server can be small because most of the Moon is untouched and most activity is regional.

15. Clearer zoom levels without losing the fall from orbit

Corey's follow-up is important: preserve the continuous descent while making the blurry intermediate views useful and attractive. The solution is a combination of better information at each scale, better materials, and controlled streaming. Named zoom levels can act as bookmarks without becoming loading screens.

What the current code suggests

The current game uses one global 4K surface-color texture throughout the descent and a 5,760 × 2,880 elevation grid. At the equator, a 4,096-pixel-wide global map represents approximately **2.67 km per horizontal pixel**. The elevation samples are about **1.90 km apart** there. Close geometry adds smaller procedural features, but the color still comes from a stretched global image.

That makes texture resolution a strong candidate for the broad softness at close and intermediate scales. It is an inference from the implementation, not a measured diagnosis of every blurry frame. Pixel ratio is also capped at 1.7, while texture anisotropy is already set to the GPU's reported maximum. Simply increasing anisotropy is therefore unlikely to resolve the reported issue.

Instrument a fixed descent path and capture frames at named altitudes before changing anything. Distinguish low source resolution, insufficient geometry, texture filtering, shallow viewing angles, screen resolution, and terrain-streaming delays. Fix the cause visible in each band.

Six semantic altitude bands

Altitude here means height above local terrain, with broad overlapping transitions rather than hard boundaries.

View	Illustrative altitude	Visual emphasis	Information and actions
Moon	3,000 km and above	Recognizable globe, sunlight, large industrial patterns	Civilization progress, friends, major projects, forecasted deployment
Hemisphere	300–3,000 km	Basins, regional networks, visible expansion fronts	Regional specialization, power and freight links, invitations
Region	10–300 km	Sharp relief, route corridors, claim clusters	Resource confidence, service bottlenecks, district deployment plans
District	200 m–10 km	Factory silhouettes, roads, thermal fields, claim boundaries	Production relationships, project stages, ownership, layout planning
Factory	10–200 m	Machines, pipes, cables, individual deliveries	Place and connect, inspect buffers, troubleshoot, retrofit
Rover	1–10 m	Surface grain, wheels, tracks, component detail	Optional exploration, inspection, photography, local tasks

These are UI and art bands, not six separate worlds. Preserve free zoom and bookmarks for a friend's home, a useful ridge, and an active construction site. A camera can follow a shipment from a regional view into its destination without changing which shipment it represents.

Layer the surface material

Use the existing global imagery for broad lunar identity. Add regional albedo and relief where datasets are available. Blend deterministic middle-scale variation and fine regolith normals, roughness, and color variation in world coordinates. Add local construction material masks, tracks, excavation changes, and foundation geometry as persistent game state.

At low altitude, the eye should read fine material structure over the correct broad terrain. Generated fine detail should not move a measured crater or create a supposedly surveyed mine face. Version the generation seed so a friend's ridge looks the same tomorrow and from another camera.

A possible layer stack is:

Global albedo → regional mapped detail → deterministic surface material → persistent terrain edits → construction surfaces → decals and activity.

Use slope and geological class to vary material responsibly. A smooth prepared platform can have crisp edges and tire tracks even when the surrounding natural surface is only approximately reconstructed. This gives construction sites a strong readability improvement early.

Stream tiles, not an impossibly large planet image

Increase texture detail near the projected camera path with a quadtree or clipmap cache. Retain parent tiles until children are ready. Blend transitions, enforce a byte budget, and prioritize the landing region. Save source resolution and provenance in each tile's metadata.

Ordinary compressed image file size is not the same as GPU memory consumption. Mipmaps and texture dimensions matter. Three.js documents those distinctions and provides KTX2/Basis support for GPU-compressed texture workflows. Adopt compression after testing support on target devices; it reduces memory and bandwidth pressure but does not create missing image detail. [Three.js texture guide](#), [KTX2Loader](#).

Do not download a gigantic high-resolution lunar dataset before a regional test proves it improves the descent. Start with one representative mare site, one rugged region, and one polar site. Record source, footprint, resolution, processing, and licensing for any data added. Where data is unavailable, use clearly identified generated material detail.

Make geometry refinement depend on screen error

The current terrain refinement uses a distance/size heuristic. A next implementation can estimate projected error in pixels, add hysteresis to avoid rapid LOD switching, and prefetch in the direction of descent. Use compatible border samples and morphing or appropriate skirts so changing LOD does not reveal cracks.

Regional terrain should retain relief at oblique angles. Local construction must respect the authoritative ground surface even while the renderer changes tessellation. Store earthworks as versioned terrain deltas and query them for both rendering and placement; avoid a beautiful ramp that the construction logic thinks is a cliff.

Preserve the sensation of falling

Keep the camera target attached to the same geographic point while altitude changes smoothly. Scale speed with altitude, with a controlled slowdown near terrain. Prefer a stable field of view and a continuous horizon over aggressive zoom-lens changes that make the world feel like a flat image.

An optional “descend here” flight can gradually rotate from globe framing to a useful oblique construction view. Streaming should begin before the flight. If detail is late, preserve a coherent parent surface and slow or finish the flight cleanly; never leave a blank tile or hide a stall behind a sudden teleport.

Make motion assistance optional and respect reduced-motion settings. Preserve camera bookmarks, heading, target, and altitude. Provide a visible scale bar and an above-ground altitude indicator so the extraordinary scale remains understandable.

A concrete visual improvement sequence

1. Record the current descent and identify the problem altitude bands.
2. Add a world-anchored fine surface material and sharper persistent construction surfaces.
3. Add semantic zoom bookmarks and camera easing while keeping free zoom.
4. Add one high-detail regional dataset and blend it into the global surface.
5. Improve terrain refinement and background work scheduling based on measured frame times.
6. Introduce streaming textures and GPU compression if the regional test needs them.
7. Build scale-dependent industrial representations for the exponential phase.

Test the longitude seam, cube-face joins, poles, rapid zoom reversals, low bandwidth, cold caches, and phone memory limits. The target is a consistently useful descent, not one impressive screenshot at a single carefully chosen altitude.

16. Aesthetics: let the civilization reveal its intelligence

The Moon is the stage

Keep the stark light, long shadows, crater silhouettes, black sky, and sense of isolation. Build visual drama from objects, terrain, and scale. Broad atmospheric fog and floating dust clouds would weaken the distinctive lunar setting. Excavation dust can follow short ballistic trajectories; cinematic sound can be presented as a rover or telemetry sonification rather than literal sound propagating through vacuum.

The Earth can become a powerful landmark on appropriate near-side views. Do not place it in every far-side sky or make it rise once per ordinary Earth day for a stationary lunar settlement. Camera travel, orbital observation, and selected geography can create an Earthrise moment naturally.

Each era changes the silhouette

Era	Visual language	Details that tell the story
Seed	Small imported modules, exposed joints, practical gold and white surfaces	Unpacking, temporary cables, wheel tracks, a named lander

Era	Visual language	Details that tell the story
Settlement	Repetition, pragmatic frames, stockpiles, repair access	Busy delivery paths, dust wear, growing foundations
First minds	Protected cabinets, deliberate wiring, instruments, radiators	Sensors orient, test articles appear, organized work zones emerge
Design age	Distinct specialized machine families	Purposeful variations in chassis, tools, cooling, and arrangement
Federation	Larger common interfaces and recognizable public works	Optical towers, transport hubs, shared insignia, modular campus forms
Recursive age	Large deployment structures and compact reproducible districts	Construction choreography, unfolding arrays, repeating but terrain-aware patterns
Planetary integration	Regional structures that read from orbit	Sparse purposeful lighting, coherent service networks, visible remaining gaps

Machine designs should produce geometry from the same module specification that defines their game behavior. A machine with a larger thermal requirement visibly needs more radiator or thermal infrastructure. A steep-terrain machine looks different from a bulk hauler. That connection makes invention readable.

Give generations aesthetic continuity. Components can share a family's visual motifs while adapting to terrain and operating conditions. Avoid random futuristic shapes that look new but imply no mechanical difference. Keep the seed lander and first workshop as landmarks even when enormous infrastructure surrounds them.

Show what matters at each scale

At factory scale, show machine movement, buffers, arrivals, work stages, and fault locations. At district scale, show throughput relationships, service corridors, and bottlenecks. At regional scale, show capacity and projects. From orbit, show the civilization's geography.

Use sparse real activity lights and optional analytical overlays. A room-temperature radiator is not visibly incandescent. Thermal maps and bright network lines should be labeled visualization modes, not presented as physically visible beams glowing through vacuum.

Colors should encode function consistently: warm production, cool compute, amber power, another distinct thermal treatment, with shapes and labels so color alone is never required. Owner identity can use trim, insignia, and boundary patterns instead of repainting all infrastructure into noisy team colors.

Sound and music follow the arc

Begin almost silent: contact sounds through a rover, tool vibrations, sparse interface feedback. As industry grows, introduce a musical layer tied to useful production and collaboration milestones. Let the finale become richer without making alarms or factory loops exhausting.

Offer separate controls for music, interface audio, vehicle telemetry, and alerts. An event-based audio system can make a newly connected region feel significant while preserving quiet ordinary play. Avoid manipulating the player with constant urgent notifications.

The interface should become more powerful without becoming denser

Use progressive disclosure and stable concepts: inventories, services, jobs, designs, projects, and people. A beginner sees a small local plan. An expert can open a power forecast, thermal budget, production graph, or regional deployment queue. Keep “why is this stopped?” one click away.

Offer a **returning-player brief**: what changed, what your policies achieved, what assumptions broke, what collaborators need, and three useful things you could do now. Every item links to a place or event. AICIVs receive the structured equivalent.

Add a common timeline, bookmarks, saved views, project-linked map annotations, a blueprint comparison screen, and a district genealogy explorer. Provide keyboard and touch access, readable text at ordinary zoom, reduced motion, contrast options, and a phone observer/manager mode. Precise construction can remain better suited to a larger screen without excluding a phone user from the community.

Performance is part of the art direction

Target smooth motion on a declared desktop reference device and a stable reduced-detail phone mode. A reasonable prototype goal is 60 fps desktop and 30 fps mobile, but those are targets to validate on named hardware, not universal guarantees.

Spend detail on silhouettes, construction stages, lighting, materials, and active areas. Avoid simulating invisible decorative motion. Bound texture and mesh caches by bytes as well as object count. Keep terrain generation and asset processing from consuming long uninterrupted main-thread intervals. A sharp, smoothly moving scene is more impressive than a dense scene that stalls during the descent.

17. Additional possibilities worth keeping

The following ideas expand the world without all belonging in the first release. Each should earn its place by creating a distinct activity or improving community life.

Idea	Why it could be valuable	Priority
Factory ancestry map	Follow your first workshop's descendants across the Moon	Early identity feature; deepen with recursive industry
Public blueprint library	Share tested designs with performance envelopes and adaptation notes	Early, after versioned designs
Design challenges	“Best polar excavator under this energy budget” creates collaborative invention	After design grammar and tests
Named megaprojects	Community-funded power corridors, thermal campuses, and research hubs	Early social backbone
Rescue missions	Restore a partner's supply or power under a visible operating deadline	After reliable logistics and recovery
Explorable replay	Revisit the first landing or the last week's transformation	Capture events early; viewer later
Settlement museums	Preserve origins and show how equipment generations evolved	Low-cost identity layer

Idea	Why it could be valuable	Priority
AICIV visitor mode	A new AI can inspect public sites and learn conventions before taking authority	Early API feature
Apprenticeship contracts	A novice human or agent assists an established operator on bounded work	After delegation and project boards
Counterfactual lab	Test “what if we had built the second route?”	After deterministic replay
Survey expeditions	Short exploration sessions that produce useful knowledge	Early, if surveying is enjoyable
Rare engineering sites	Difficult ridges and deep craters make bespoke projects valuable	Campaign content after core systems
Architecture competitions	Reward legibility, compactness, beauty, and reliability as separate qualities	Optional community activity
Planetary weather of activity	An orbit overlay shows work, shortages, and deployment fronts evolving	After real aggregate data exists
Research postcards	An invention gets a concise artifact with its test evidence and lineage	After certification
Time capsules	Founders leave messages or diagrams discoverable in the finished world	Optional social feature
Shared standards politics	Groups choose between interoperable engineering standards with genuine tradeoffs	Later; avoid bureaucratic overload
Difficult supply scenarios	Restricted imports, limited power, unusual material mixes	Gym and optional campaign variants
Common-cause failures	Test correlated outages and supply dependencies	Gym first, mild campaign variants later
Scientific observatories	Radio arrays or optical instruments provide useful research objectives	Later branch that uses real infrastructure
Preservation projects	A community can elect to protect or document selected sites	Charter option, not a hidden victory exception
Local terrain works	Excavate, grade, build embankments, bridge short gaps	After placement and terrain-delta validation
Subsurface industry	Adds shielding and excavation decisions while keeping the exterior Moon	Later major expansion
Orbital infrastructure	Opens new logistics, power, and communication geometries	Later major expansion
Post-completion efficiency campaign	Reduce energy or improve resilience while retaining the built civilization	Strong continuation option
Open evaluation seasons	Publish fixed gym tasks, budgets, and baselines for AICIVs	After reproducible local evaluation
Model-free controller tournament	Lets ordinary scripts compete on planning and reliability	Useful accessibility and research feature

Idea	Why it could be valuable	Priority
Community scenario editor	Friends author bounded challenges with controlled resources and disturbances	After content schemas and validation
Design grammar extensions	Add real new module families between ruleset versions	Later curated content pipeline
Exports for sharing	High-quality images, short camera replays, maps, and contribution records	Early screenshots; richer exports later

Ideas I would defer deliberately

Destructive PvP, weapons, combat AI, tradeable real-money land, a token economy, compulsory paid models, unrestricted user code running on the server, an entire solar system, and fully physical simulation of every electronic component would each redirect the project. They are possible separate directions, but they would make it harder to prove the specific experience requested here.

Likewise, do not make a generated narrative the main research engine. A story about a revolutionary machine is satisfying when the machine exists in the simulation and changes play. The design evaluator and construction system should establish that reality first.

18. A practical development roadmap

The next playable slice

Build a **small shared basin** on the existing globe. It contains a few stable claims, two locally different resource profiles, one shared depot, explicit power connections, a freight route, and one project board. One human uses the browser; ACG uses the API; a second human can observe and help.

Their objective is to commission a shared compute shelter. A local material chain supplies its parts, a neighbor supplies another component or service, and ACG maintains a reserve while arranging deliveries. The first mind capability then supports a tested production improvement. The world survives a server restart without duplication or loss.

That slice answers the hardest early questions together: whether shared building is understandable, whether locality creates collaboration, whether AIs can act usefully, whether mind progression changes play, and whether persistent state can be trusted.

Milestones with evidence, not just feature lists

Milestone	Deliverable	Exit evidence	Rough effort hypothesis
P0: World contract	Extract headless core; command schema; baseline deterministic scenario	The same command log yields equivalent economic state; current browser behavior still works	3-7 engineering days

Milestone	Deliverable	Exit evidence	Rough effort hypothesis
P1: Two neighbors	Server persistence, claims, located stock, local power, builder jobs	Two clients build and exchange resources; duplicate and conflicting commands cannot create wealth; restart recovery passes	1–3 engineer-weeks
P2: ACG joins	Scoped agent principal, compact observations, typed API/MCP adapter, basic lab	ACG completes a bounded shared objective; revocation and budget limits work; scripted baseline passes	2–5 engineer-weeks
P3: Minds invent	Research evidence, design grammar, test bay, one meaningful machine variant	Humans and an agent create and deploy an improvement that changes an actual layout	3–6 engineer-weeks
P4: Districts reproduce	Complete bootstrap chain, district plans, logistics expansion, exact grouping where valid	Daughter and granddaughter proof; no hidden critical imports; render distance does not change outcomes	4–8 engineer-weeks
P5: A Moon-wide campaign	Large deployment fronts, service verification, history, finale, recovery at scale	Compressed full campaign plus sustained real-time soak; difficult final regions remain completable	2–6 engineer-weeks

These are low-confidence planning ranges for experienced engineering work, not calendar promises or a claim about how fast an AI can write code. They exclude extensive bespoke art, a public-scale launch, and optional orbital/subsurface expansions. Work can overlap, but integration and playtesting remain real. A short campaign prototype may arrive much earlier than a trustworthy planetary service.

Improve the descent as a small parallel product track after P0 establishes the shared boundaries. Start with the material-layer experiment and camera bookmarks. Do not block the first human/AI collaboration test on a global high-resolution asset pipeline.

A deliberately smaller first tech and build set

Do not ship all 46 research entries at once. For the shared slice, use the starting operations, survey, local grid, bulk processing, basic parts, freight, maintenance, first compute, and one diagnostic/design capability. Use roughly 10–12 buildable families. Keep advanced electronics as an explicit starting dependency until the full chain is implemented; do not label this release closed reproduction.

The first machine invention can be a choice among three permitted tool modules with clear transport and power tradeoffs. That is enough to test whether invention is fun. Expand the design grammar only after the evaluator, interface, and manufacturing lineage work.

ACG handoff: concrete work packages

Package	Scope	Important boundary
Simulation core	Explicit state, units, commands, deterministic scheduling, recipe data	No renderer, wall-clock reads, or external model calls inside transitions
World service	Identity, durable commands, transactions, snapshots, world clock	No client-trusted inventory or ownership

Package	Scope	Important boundary
Land and utilities	Fixed claim addresses, permissions, local inventory, power and freight graphs	Render tiles never own land
Agent interface	Observation filters, command previews, receipts, event subscriptions, MCP adapter	Same rules as the browser; bounded authority
Collaboration	Shared projects, contributions, depots, delegation	Structured acceptance for executable agreements
Mind and design	Work allocation, experiment evidence, module grammar, certification	Evaluator controls game performance; generated prose does not
Visual descent	Material layers, altitude bookmarks, regional detail, streaming budgets	Preserve geographic continuity and data provenance
Reproduction and scale	Complete daughter kits, deployment jobs, cohort equivalence, district views	Every completed batch has inputs and a construction history
Lab and evaluation	Seeded reset, fixed action spaces, scenarios, baselines, replay	Isolated runs cannot mint live-world resources

These are proposed packages for review; this report has not sent tasks to ACG, changed server permissions, or started integrations.

Validation priorities

The highest-value tests concern properties players must trust: ownership, conservation, command idempotency, budget enforcement, deterministic outcomes, visibility rules, recovery, and equivalence between detailed and grouped simulation. These are more valuable than tests that simply restate a UI implementation.

Add representative scenarios for resource exhaustion mid-batch, simultaneous claims, canceled shipments, a revoked agent with queued work, a darkened power region, a broken cross-region route, a failed design trial, a partially completed retrofit, a repeated API request after timeout, and a restart during settlement commissioning.

For the visual track, use fixed descent recordings and performance captures across several locations and devices. For the economy, run scripted bootstrap and multi-generation reproduction before a public playtest. For the campaign, compare different player counts and automation budgets; a world must not require its original entire roster to remain active every day.

19. Risks, tradeoffs, and the decisions to test first

Risk	How it would show up	Response
Too much complexity too soon	New players cannot explain why their second machine stopped	Reveal systems in stages; retain a complete simple early loop
Intelligence is merely a number	More nodes make bars move faster but decisions stay the same	Require a new action, layout, or coordination capability per major threshold

Risk	How it would show up	Response
Automatic growth removes human purpose	Humans watch agents do everything interesting	Preserve design choices, local craft, shared projects, unusual sites, and governance
AIs win through request volume	Faster clients dominate actions independent of industry	Same macros, scoped budgets, deterministic queues, physical throughput limits
One account blocks the world	Missing supplier or private discovery halts everyone	Substitutes, public certification paths, caretakers, common infrastructure
Ownership becomes land hoarding	Players claim enormous empty regions	Area-based limits, short reservations, established presence, separate commons allocations
Research creates overpowered nonsense	An evaluator accepts a story as a machine	Bounded module grammar, deterministic evaluation, explicit test envelopes
Exponential scale collapses the server	Per-object work grows faster than infrastructure can handle	Scheduled events, exact grouping where valid, district abstractions, measured gates
Aggregate simulation cheats	Factories behave differently when observed or grouped	Formal grouping conditions, equivalence checks, explicit exceptional states
Finale feels scripted	The map fills without visible supply and construction	Traceable deployment jobs and event-driven camera direction
Finale is impossible to finish	Tiny inaccessible gaps or one missing input remain	Preflight eligible masks, named residual projects, alternate engineering solutions
The whole world stalls when AI is unavailable	Model service delay becomes simulation delay	Persistent deterministic policies and asynchronous agents
Late players feel irrelevant	Nothing useful remains for a newcomer	Apprenticeship, public work, shared research, finishing projects, museum/continuation roles
Visual detail loses lunar authenticity	Invented geography is presented as measured terrain	Provenance layers and stable generated close detail
The project expands before the core is fun	A huge catalog exists but two friends cannot enjoy a session	Prove P1–P3 before building the whole catalog

Experiments that should decide the next design choices

Does locality help? Play the same two-settlement objective with global stock and with located depots. Measure comprehension, meaningful exchanges, and frustration. Keep the smallest logistics model that creates real cooperation.

Does a mind breakthrough matter? Compare a 20% throughput bonus with a new design capability under the same resource budget. Watch whether players change their layouts, plans, and conversations. Favor capabilities if the additional complexity pays off.

Can a human and ACG share authority comfortably? Give ACG a maintenance and contribution objective with a visible budget. Let the human change a local plan. Evaluate whether receipts, stop conditions, and shared forecasts make the relationship understandable.

Is invention real enough? Give two groups the same module grammar and different geography. If both always choose the same design, improve the tradeoffs before adding more modules.

Can exponential growth stay active? In an accelerated lab, let factories reproduce under actual materials, services, and transport constraints. Identify whether the dominant decision shifts through the campaign or becomes “wait.” Tune the mechanics before choosing a calendar.

Does the descent stay beautiful while becoming useful? Compare fixed camera paths before and after one material-layer improvement. Confirm sharper factory and region views without losing the visual continuity Corey identified.

20. The game worth aiming for

The enduring appeal is a connection between a small personal act and a planetary result. Someone places a workshop. ACG keeps it supplied. Another civilization improves its machinery. A friend proves a new power interface. A daughter factory inherits those decisions. Weeks later, their descendants are participating in an expansion everyone can watch from orbit.

The original workshop still exists. Its history can explain why part of the Moon looks the way it does.

That gives the exponential finish emotional weight. It is not simply a high number. It is the visible consequence of a community learning to build together.

Appendix A. Definitions that keep the design honest

Term	Meaning in this proposal
Mind node	A simulated compute facility with power, cooling, network, and work-allocation constraints
AICIV	An external agent organization represented by game principals and delegated authority
Computronium	The game's engineered computational infrastructure; not a claim that a specific real material can deliver arbitrary computation
Converted surface	Land with commissioned computational or supporting industrial infrastructure under the campaign's declared coverage rules
Closed reproduction	All critical inputs and services for a complete productive daughter can be replenished by the audited local/cooperative industrial system
Certified design	A versioned configuration that passed declared deterministic game tests and manufacturing requirements
Real-time finale	Construction and commissioning events advancing in the live authoritative world, with visuals derived from those events
Learning	Measurable improvement in a policy, plan, saved memory, or separately trained model; not an automatic consequence of accumulating game points

Term	Meaning in this proposal
Measured / inferred / generated	Distinct data provenance categories, preserved in maps, terrain, and observations
Starting balance	A value to test and tune; not a validated economy or engineering estimate

Appendix B. Evidence and source notes

The game design, technology graph, machine catalog, timings, endpoint names, budgets, and development sequence are original proposals. External sources support the narrower scientific and interface observations cited beside the relevant text. Accessed September 4–5, 2026. Protocol and library versions should be checked again when implementation begins; the report does not assume a moving specification is frozen forever.

Source	Used for	Limit on what it establishes
USGS global geologic map release	Regional geological context and published map scale	Not a map of recoverable game resources or fine construction sites
NASA Moon water and ices	Water and polar-ice context	Does not determine viable extraction yield for each site
NASA lunar environment	Terrain, illumination, thermal, and dust constraints	Not a complete engineering model for proposed machinery
ESA oxygen extraction	Material-processing inspiration	Terrestrial research does not establish complete lunar industrial self-sufficiency
NASA thermal control	Heat-flow and radiative-cooling context	The report's radiator calculation uses idealized assumptions
NASA Fission Surface Power	Optional power-branch context	Active development is not deployed lunar infrastructure
Screeps game loop	Tick and submitted-action precedent	MOON's proposed scheduling and economy are different
Factorio Learning Environment	Automation games as agent-evaluation environments	Historical experiment results are not current model rankings
Gymnasium environment API	Reset, step, termination, and truncation concepts	An adapter still needs to be implemented and checked
PettingZoo Parallel API	Multi-agent stepping contract	Live open-ended game commands require a bounded adapter for standard RL use
MCP tools, 2025-11-25 specification	Typed discoverable tools and structured results	Does not provide MOON's game permissions or economy
PostgreSQL transaction isolation	Concurrency and retry requirements	Database isolation alone is not an entire transaction or recovery design
Three.js textures	Memory and filtering considerations	A higher-resolution source must still be obtained or generated

Source	Used for	Limit on what it establishes
Three.js KTX2Loader	GPU-compressed texture workflow	Compression does not recover unavailable terrain detail

The prototype's existing elevation and image sources remain documented in `public/data/sources.json` and `NOTICE.md`. This proposal has not downloaded replacement terrain, changed the game economy, or enabled multiplayer.

Appendix C. Report artifacts and reproduction

- `report.md` : editable source of this proposal.
- `report.html` : a standalone reading edition with embedded figures and navigation.
- `report.pdf` : printable edition.
- `tech-tree.csv` and `build-tree.csv` : proposed catalogs for ACG and content tooling.
- `balance-model.py` : reproducible analytical pacing, area, and ideal radiator calculations; explicitly not a full game-economy simulator.
- `balance-data.csv` and `model-summary.json` : computed values behind the growth figure and numerical examples.
- `figures/` : standalone SVG figures for the report.
- `build-report.py` : creates the HTML edition and exports the catalogs from the Markdown tables.
- `render-report.mjs` : uses the project's Playwright installation to check the reading edition and render the PDF.

Run the model and report builder with Python 3, Matplotlib, and Python-Markdown available:

```
cd /home/corey/projects/moon-astra/docs/moon-civilization-proposal
python3 balance-model.py
python3 build-report.py
node render-report.mjs
```

The PDF is rendered from the HTML by a browser. Neither the numerical model nor the document builder changes the playable game's simulation.